FLEET OPERATIONS AND VEHICLE ALLOCATION SYSTEM

A PROJECT REPORT

Submitted by

ANANYA P

2116241401010

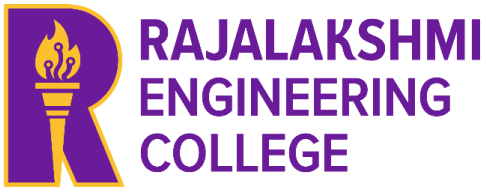
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CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled **FLEET OPERATIONS AND VEHICLE ALLOCATION SYSTEM** is the bonafide work of **ANANYA P (241401010)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

SIGNATURE

Dr. K. R. Sowmia, Ph.D.,
HEAD OF THE DEPARTMENT
Department of Computer Science and Business Systems
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

M. L. Alphin Ezhil Manuel,
COURSE FACULTY
Assistant Professor (SG)
Department of Computer Science and Business Systems
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on 03/11/2026.

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, **ANANYA P (2116241401010)**, hereby declare that the project report titled "**FLEET OPERATIONS AND VEHICLE ALLOCATION SYSTEM**", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report, including its data model, forms, reports, pages, Deluge scripts and workflows, was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every external resource and code reference used in this work has been duly acknowledged in the report.

Place: Chennai

SIGNATURE OF THE STUDENT

Date: 03/11/2026

ANANYA P
241401010

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ANANYA P

ABSTRACT

Fleet operations involve the coordination of vehicles, drivers, delivery requests and trip-related expenses. In logistics operations, managing these activities manually or through separate spreadsheets can make it difficult to identify available resources, assign suitable vehicles, monitor delivery status and maintain accurate trip-cost records. Manual allocation may also result in delays, inconsistent records and inefficient utilisation of vehicles and drivers. To address these challenges, the Fleet Operations and Vehicle Allocation System was developed using Zoho Creator, a low-code application development platform.

The application provides a centralised system for maintaining vehicle, driver, delivery request and trip information. Four primary forms—Vehicles, Drivers, Delivery_Requests and Trips—are used to capture and organise operational data. Lookup fields connect trip records with their corresponding delivery requests, vehicles and drivers. The Trips form also contains Fuel_Details and Additional_Expenses subforms to record fuel purchases and other trip-related expenses. Deluge workflows automate vehicle and driver allocation by checking resource availability and vehicle load capacity against the delivery request. Additional automation calculates total trip cost, fuel efficiency and cost per kilometre. Reports such as All_Vehicles, All_Drivers, Pending_Requests and Trips provide access to operational records. The Fleet_Dashboard page presents key metrics and charts related to vehicle availability, pending deliveries and completed trips. The application also defines Admin_Access, Manager_Access and Driver_Access permission profiles to organise access according to user responsibilities.

The application was tested using sample vehicle, driver, delivery and trip records. Allocation was verified using a delivery request, and the cost and efficiency calculations were checked using sample trip values. For example, a trip with 200 kilometres travelled, 20 litres of fuel consumed and a total cost of ₹2,300 produces a fuel efficiency of 10 km/L and a cost of ₹11.50 per kilometre. The completed application was published on Zoho Creator. By bringing operational records, resource allocation and trip calculations into one application, the system reduces repetitive manual work, improves consistency of records and supports more organised fleet management.

PROJECT AT A GLANCE

Attribute	Details
Project Title	FLEET OPERATIONS AND VEHICLE ALLOCATION SYSTEM
Application Domain	Logistics and Fleet Management
Student Name & Register No.	ANANYA P – 2116241401010
Zoho Creator Edition	Trial Edition
Application Name (Link Name)	Fleet Operations and Vehicle Allocation System
Published / Live URL	https://ananyap2024csbsrajala436.zohocreatorportal.in
Demo Credentials (per role)	Demo Credentials: Username – *ananya*, Password - *kKVVvc959zXLE7D@* (Customer role).
Number of Forms / Subforms	4 Forms / 2 Subforms
Number of Reports (by type)	List: 4 Kanban: 0 Calendar: 0 Pivot/Chart: 2 Others: 0
Number of Pages / Dashboards	1
Number of Workflows	Form: 5 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	Deluge workflow scripts for automatic resource allocation, trip cost calculation, fuel efficiency calculation and cost-per-kilometre calculation.
Integrations	Zoho Creator forms, lookup fields, reports, pages and Deluge workflows.
Roles & Profiles Defined	Admin_Access, Manager_Access, Driver_Access
Environments Used	Zoho Creator application development and published application.
Date of Production Deployment	01 November 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

1.1 Background

Fleet management is an important part of logistics and transportation operations. It involves managing vehicles, drivers, delivery requests, trip schedules and operational expenses. Organisations that handle multiple deliveries must ensure that suitable vehicles and drivers are available for each request while maintaining accurate records of their operations.

When these activities are handled manually or through independent spreadsheets, information may become scattered across different records. Identifying available vehicles, checking their load capacity, assigning drivers and calculating trip expenses can require repeated manual effort. This may result in delays, inconsistent information and difficulty in monitoring the overall status of fleet operations.

The Fleet Operations and Vehicle Allocation System was developed to address these challenges through a centralised application. The application uses Zoho Creator to maintain fleet-related information, automate resource allocation and calculate important trip metrics. It provides a structured approach to managing delivery requests and monitoring vehicle and driver availability.

1.2 Problem Statement

1.2 Problem Statement

- Managing fleet operations requires coordination between vehicles, drivers and delivery requests. In a manual system, the availability and capacity of vehicles may need to be checked separately before assigning a delivery. Driver availability must also be verified, and trip-related expenses must be recorded and calculated independently.
- These processes can lead to the following problems:
- 1. Difficulty in maintaining centralised vehicle and driver records.

1.3 Objectives

- 2. Time-consuming identification of suitable vehicles based on availability and load capacity.
- 3. Manual coordination of drivers and delivery requests.
- 4. Possibility of inconsistent or outdated resource availability information.
- 5. Repetitive calculation of fuel efficiency, total trip cost and cost per kilometre.
- 6. Difficulty in obtaining a consolidated view of pending deliveries, available vehicles and completed trips.
- Therefore, a centralised system is required to organise fleet records, automate basic allocation activities and calculate trip-related operational metrics.

1.4 Scope of the Project

1.3 Objectives

- The primary objective of this project is to develop a Fleet Operations and Vehicle Allocation System using Zoho Creator.
- The specific objectives are:
- 1. To create a centralised database for storing vehicle, driver, delivery request and trip information.
- 2. To automate vehicle allocation by checking vehicle availability and load capacity.
- 3. To assign available drivers to delivery requests through workflow automation.
- 4. To maintain relationships between delivery requests, vehicles, drivers and trips using lookup fields.
- 5. To record fuel details and additional trip expenses through subforms.
- 6. To calculate total trip cost, fuel efficiency and cost per kilometre using Deluge scripts.

7. To generate reports for viewing vehicle, driver, delivery and trip information.

- 8. To develop a dashboard displaying important fleet operation metrics and charts.

- 9. To configure role-based permissions for administrators, managers and drivers.
- 1.4 Scope of the Project
- The project focuses on the digital management of basic fleet operations through a Zoho Creator application. It provides features for maintaining vehicle and driver records, registering delivery requests, allocating available resources and recording trip details.

1.5 Justification for a Low-Code Approach

The system includes vehicle information such as vehicle type, load capacity, mileage and availability. Driver information includes name, phone number, licence number and availability. Delivery requests contain the source, destination, load weight, priority and request status.

The application creates trip records by associating a delivery request with a suitable available vehicle and driver. Vehicle selection is based on the availability status and the capacity required for the delivery. Trip records also maintain distance, fuel usage, fuel details and additional expenses.

The application calculates total trip cost, fuel efficiency and cost per kilometre. Reports and a dashboard provide summarised operational information. Role-based permission profiles are configured for administrators, managers and drivers.

The project is intended as an academic fleet-management prototype. It focuses on record management, rule-based allocation and operational calculations within Zoho Creator.

1.6 Organisation of the Report

1.5 Justification for a Low-Code Approach

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

2.1 Study of the Existing System

In a conventional fleet-management process, vehicle details, driver information and delivery requests may be maintained using separate spreadsheets, registers or independent records. When a new delivery request is received, the operator must examine the available vehicles and identify one with sufficient load capacity. Driver availability must then be checked before the assignment is recorded.

Trip details such as distance travelled, fuel consumed and additional expenses may be maintained separately. Calculating fuel efficiency and cost per kilometre requires manual computation or spreadsheet formulas. The overall status of fleet operations may be obtained by combining information from multiple records.

Although this approach can be used for a small number of deliveries, it becomes difficult to maintain consistency when the number of vehicles, drivers and delivery requests increases.

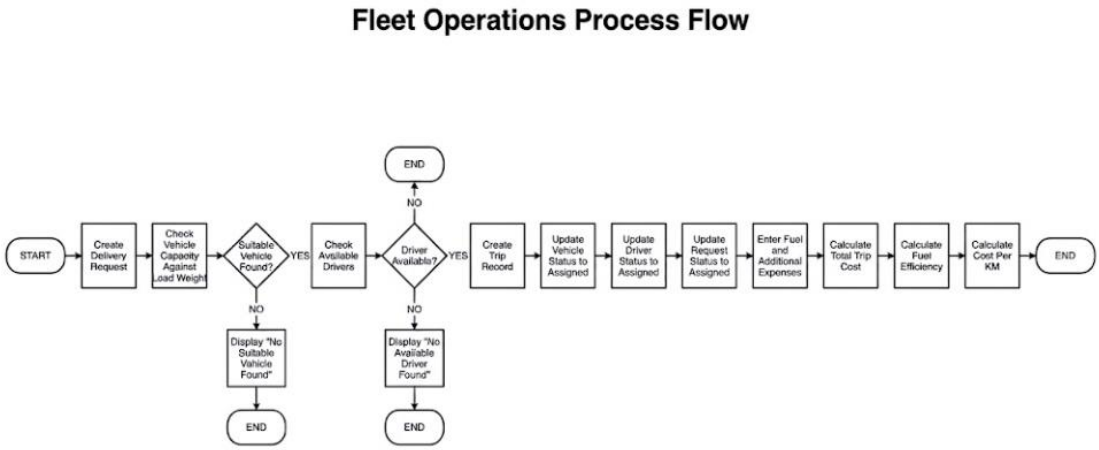


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- The existing manual or spreadsheet-based approach has several limitations:
- 1. Scattered information: Vehicle, driver, delivery and trip information may be stored in separate files or registers.

- 2. Manual allocation: Operators must manually check vehicle capacity and availability before assigning a delivery.

2.3 Low-Code versus Traditional Development

3. Resource coordination: Driver availability must be checked separately, increasing the effort required to complete an assignment.

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Approximately 8–10 weeks	Approximately 2–3 weeks
Skills required	Programming and database knowledge	Basic technical knowledge
Hosting & infrastructure	Must be configured	Platform-managed hosting
Mobile application	Requires additional development	Supported mobile access
Security & access control	Must be implemented	Built-in access permissions
Maintenance & change	Requires code modifications	Can be updated through visual tools
Customisation limits	High flexibility	Depends on platform capabilities
Cost	May require more development resources	Can reduce development effort; subscription charges may apply

4. Repetitive calculations: Fuel efficiency, total trip cost and cost per kilometre may need to be calculated manually.

2.4 Literature Survey

5. Delayed status updates: Vehicle, driver and delivery statuses may not be updated consistently across different records.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Richardson (2014)	Low-Code Platforms	Faster development	Saves time
2	Gartner (2021)	Low-Code Applications	Simplifies development	Easy building
3	Zoho Creator Documentation	Forms and reports	App development	—
4	Zoho Deluge Documentation	Workflow automation	Auto calculations	—
5	OWASP Security Standards	Data protection	Secure access	—

6. Limited operational visibility: Obtaining a consolidated view of pending deliveries, available vehicles and completed trips may require additional manual effort.

2.5 Emerging Trends in Low-Code Development

7. Record maintenance: Maintaining consistent and structured records becomes more difficult as operational data increases.

1. **2.3 Low-Code versus Traditional Development**
2. **Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application**
3. **Criterion: Development time**
4. **Traditional (Pro-Code) Development: Requires separate development of the user interface, database structure, business logic, access control and deployment components.**
5. **Low-Code on Zoho Creator: Forms, reports, pages and workflows can be configured through the platform, reducing the amount of application infrastructure that must be developed manually.**

Criterion: Skills required

2.6 Proposed System

Traditional (Pro-Code) Development: Requires programming knowledge, database development, front-end development, back-end development and deployment knowledge.

Low-Code on Zoho Creator: Requires knowledge of application design, data modelling, Zoho Creator configuration and Deluge scripting for customised business logic.

Criterion: Hosting & infrastructure

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Application owner	Manage application records, configure access permissions, and monitor all fleet operations.
Manager	Fleet operations manager	View and update vehicle, driver, delivery request and trip records; monitor operational reports and dashboard information.
Driver	Driver using the application	View and update permitted trip records and access relevant fleet information according to assigned permissions.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow authorised users to add and maintain vehicle records with mandatory details such as vehicle ID, type and capacity.	Administrator, Manager	High
FR-02	The system shall allow authorised users to register drivers with their contact details, licence number and availability.	Administrator, Manager	High
FR-03	The system shall allow users to create delivery requests containing source, destination, load weight and priority.	Administrator, Manager	High
FR-04	The system shall automatically identify an available vehicle whose load capacity is sufficient for a delivery request.	System	High
FR-05	The system shall check driver availability and create a trip record when a suitable vehicle and driver are available.	System	High
FR-06	The system shall calculate total trip cost using fuel details and additional expenses.	System	High
FR-07	The system shall calculate fuel efficiency and cost per kilometre using trip distance, fuel consumption and total cost.	System	High
FR-08	The system shall display fleet operation information through reports and a dashboard, including vehicle availability, pending deliveries and completed trips.	Administrator, Manager	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	The application shall provide access to fleet records and reports through the Zoho Creator platform under normal operating conditions.
NFR-02	Security	Users shall access application components according to the permissions assigned to their respective profiles.
NFR-03	Usability	The application shall provide structured forms, reports and dashboard components to support easy navigation and data entry.
NFR-04	Availability	The application shall be accessible through the published Zoho Creator environment when the platform and network connection are available.
NFR-05	Maintainability	Forms, reports, pages and workflows shall be configurable and maintainable through the Zoho Creator development environment.
NFR-06	Auditability	Records shall be maintained in a structured manner, and available platform record-history or audit features may be used to trace changes where enabled.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Trial Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not required; the application does not depend on external API calls.
Other Zoho services	Not required; the application is implemented using Zoho Creator components.
Mobile	Zoho Creator mobile application (optional access, subject to device and platform support).
Client hardware	Computer or laptop with Intel Core i3 or equivalent processor, 4 GB RAM or above, and an internet connection.

3.5 Use Case Model

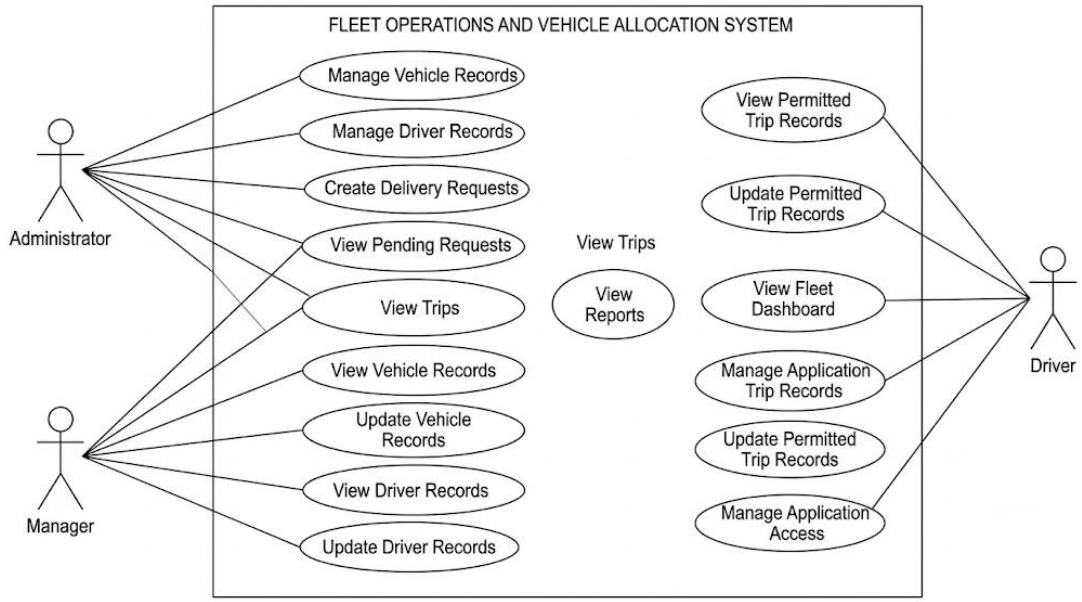


Figure 3.1: Use Case Diagram of the Fleet Operations and Vehicle Allocation System

Table 3.5: Use Case Description – Automatic Vehicle and Driver Allocation

Use Case ID	UC-01
Use Case Name	Automatic Vehicle and Driver Allocation
Actor(s)	Administrator / Manager / System
Pre-condition	A delivery request has been entered with a valid load weight. At least one vehicle and one driver must be available for successful allocation.
Main Flow	1. The user creates and submits a delivery request. 2. The system retrieves vehicles marked as Available. 3. The system checks whether the vehicle capacity is greater than or equal to the requested load weight. 4. The system retrieves drivers marked as Available. 5. If both suitable resources are found, the system creates a trip record. 6. The selected vehicle and driver availability statuses are updated to Assigned. 7. The delivery request status is updated to Assigned.
Alternate Flow	If no suitable vehicle is available, the system displays a message indicating that no suitable vehicle was found. If no available driver is found, the system displays a corresponding message. The request is not automatically allocated in these cases.
Post-condition	A trip record is created and the corresponding delivery request, vehicle and driver are associated with the allocation.
Related FR	FR-03, FR-04, FR-05

Table 3.6: Use Case Description – Trip Expense and Metric Calculation

Use Case ID	UC-02
Use Case Name	Trip Expense and Metric Calculation
Actor(s)	Administrator / Manager / System
Pre-condition	A trip record exists and the required trip distance, fuel usage and expense details are entered.
Main Flow	1. The user enters fuel details in the Fuel_Details subform. 2. The user enters additional expenses in the Additional_Expenses subform. 3. The system calculates the total trip cost by adding the recorded fuel and additional expenses. 4. The user enters or updates the distance travelled and fuel used. 5. The system calculates fuel efficiency using distance divided by fuel used. 6. The system calculates cost per kilometre using total cost divided by distance.
Alternate Flow	If fuel usage or distance is zero or invalid, the corresponding calculation is not performed and a validation message is displayed where configured.
Post-condition	The trip record contains the calculated total cost, fuel efficiency and cost per kilometre when valid input values are available.
Related FR	FR-06, FR-07

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

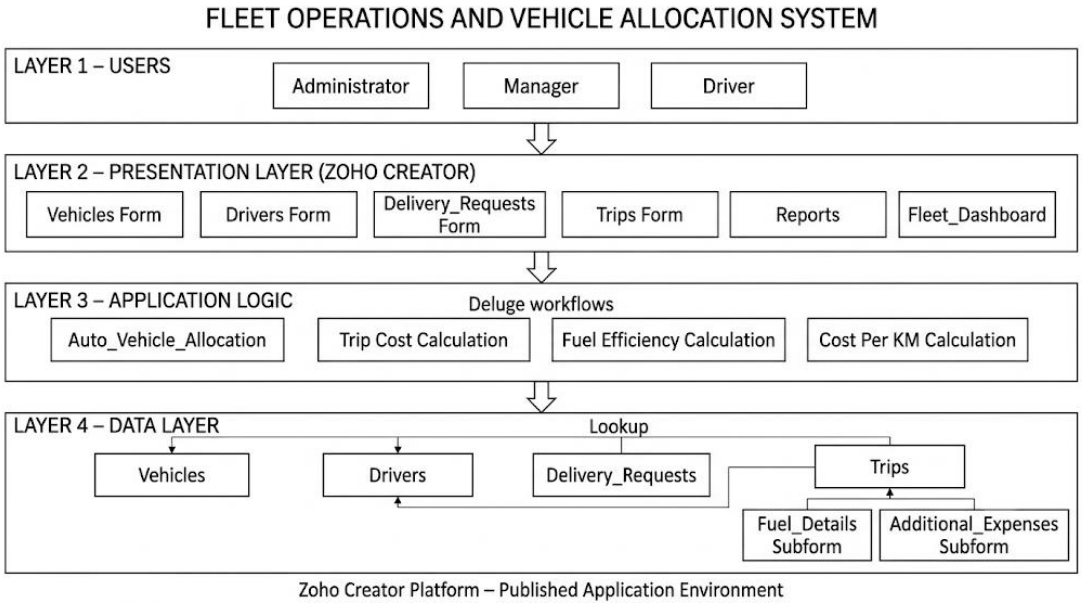


Figure 4.1: System Architecture of the Fleet Operations and Vehicle Allocation System

The Fleet Operations and Vehicle Allocation System follows a layered application architecture implemented using Zoho Creator.
The **user layer** consists of three application roles: Administrator, Manager and Driver. Each role accesses the application according to the permissions assigned through the corresponding permission profile
.
The **presentation layer** consists of Zoho Creator forms, reports and the Fleet_Dashboard page. Forms are used to capture vehicle, driver, delivery request and trip information. Reports provide structured access to stored records, while the dashboard presents important fleet operation metrics and charts.

The **application logic layer** consists of Deluge scripts and form workflows. These workflows perform automatic vehicle and driver allocation, create trip records and calculate trip-related values such as total cost, fuel efficiency and cost per kilometre.

The **data layer** contains the Vehicles, Drivers, Delivery_Requests and Trips forms. Lookup fields establish relationships between trips, delivery requests, vehicles and drivers. The Trips form also contains Fuel_Details and Additional_Expenses subforms to maintain related expense information.

The application is developed and published within the Zoho Creator environment. The platform provides the application hosting and execution environment for the configured forms, reports, pages and workflows.

4.2 Data Model

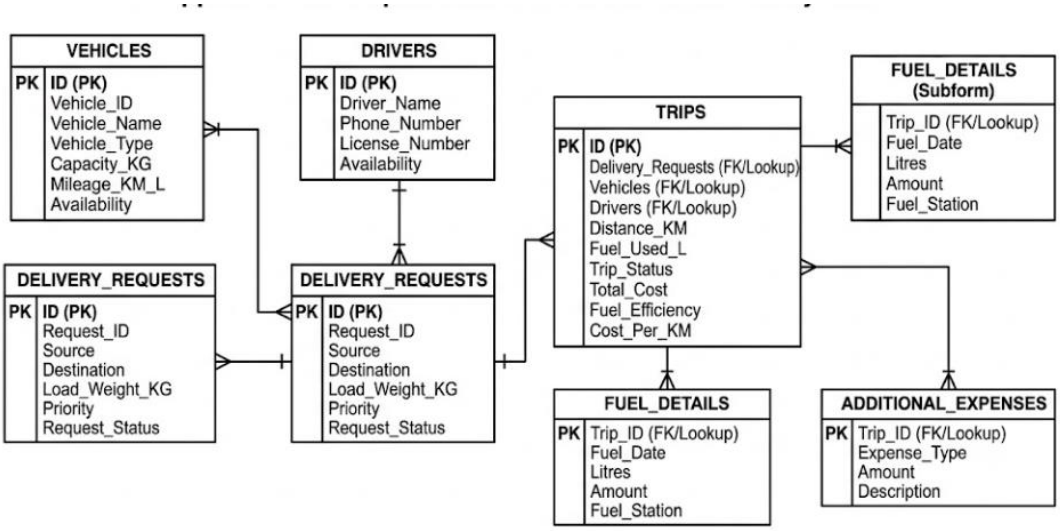


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Vehicles	Vehicle_ID, Vehicle_Name, Vehicle_Type, Capacity_KG, Availability	Trips	One-to-Many	Lookup in Trips
Drivers	Driver_Name, Phone_Number, License_Number, Availability	Trips	One-to-Many	Lookup in Trips
Delivery_Requests	Request_ID, Source, Destination, Load_Weight_KG, Priority, Request_Status	Trips	One-to-One allocation relationship	Lookup in Trips
Trips	Distance_KM, Fuel_Used_L, Trip_Status, Total_Cost, Fuel_Efficiency, Cost_Per_KM	Fuel_Details; Additional_Expenses	Parent–Child	Subforms
Fuel_Details	Fuel_Date, Litres, Amount, Fuel_Station	Trips	Parent–Child	Subform in Trips
Additional_Expenses	Expense_Type, Amount, Description	Trips	Parent–Child	Subform in Trips

4.3 Form Design

The application consists of four primary forms: Vehicles, Drivers, Delivery_Requests and Trips. These forms capture the information required to manage fleet resources and delivery operations.

The Vehicles form stores the vehicle ID, vehicle name, vehicle type, load capacity, mileage and availability. The vehicle type and availability fields use dropdown selections to maintain consistent values.
The Drivers form stores driver name, phone number, licence number and availability. The availability field helps the allocation workflow identify drivers who can be assigned to delivery requests.
The Delivery_Requests form captures the request ID, source, destination, load weight, priority and request status. The priority field allows requests to be classified as High, Medium or Low. The request status indicates whether a request is Pending, Assigned or Completed.
The Trips form connects a delivery request with its allocated vehicle and driver using lookup fields. It also stores distance travelled, fuel used, trip status, total cost, fuel efficiency and cost per kilometre.
Two subforms are included within the Trips form. Fuel_Details records the date of fuel purchase, litres, amount and fuel station. Additional_Expenses records expense type, amount and description.
The field specifications are provided in the following tables.

Table 4.2: Field Specification – Vehicles and Drivers Forms

VEHICLES FORM					
Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Vehicle ID	Vehicle_ID	Single Line	Yes	Required	Unique vehicle identifier
Vehicle Name	Vehicle_Name	Single Line	Yes	Required	Readable vehicle name
Vehicle Type	Vehicle_Type	Drop Down	Yes	Van / Mini Truck / Truck	Category of vehicle
Capacity (KG)	Capacity_KG	Number	Yes	Positive numeric value	Maximum load capacity in kilograms
Mileage (KM/L)	Mileage_KM_L	Decimal	No	Numeric value	Vehicle fuel mileage
Availability	Availability	Drop Down	Yes	Available / Assigned / Maintenance	Current vehicle status

DRIVERS FORM					
Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Driver Name	Driver_Name	Single Line	Yes	Required	Name of the driver
Phone Number	Phone_Number	Phone	Yes	Valid phone number	Driver contact information
License Number	License_Number	Single Line	Yes	Required	Driver licence identification
Availability	Availability	Drop Down	Yes	Available / Assigned / Unavailable	Current driver status

Table 4.3: Field Specification – Delivery Request and Trip Forms

DELIVERY_REQUESTS FORM					
Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Request ID	Request_ID	Single Line	Yes	Required	Unique delivery request identifier
Source	Source	Single Line	Yes	Required	Delivery origin
Destination	Destination	Single Line	Yes	Required	Delivery destination
Load Weight (KG)	Load_Weight_KG	Number	Yes	Positive numeric value	Required delivery load weight
Priority	Priority	Drop Down	No	High / Medium / Low; default Medium	Delivery priority
Request Status	Request_Status	Drop Down	Yes	Pending / Assigned / Completed; default Pending	Current delivery request status

TRIPS FORM					
Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Delivery Request	Delivery_Requests	Lookup	Yes	Select from Delivery_Requests	Associated delivery request
Vehicle	Vehicles	Lookup	Yes	Select from Vehicles	Vehicle allocated to the trip
Driver	Drivers	Lookup	Yes	Select from Drivers	Driver assigned to the trip
Distance (KM)	Distance_KM	Decimal	No	Positive numeric value	Distance travelled
Fuel Used (L)	Fuel_Used_L	Decimal	No	Positive numeric value	Total fuel consumed
Trip Status	Trip_Status	Drop Down	Yes	Assigned / In Progress / Completed; default Assigned	Current trip status
Total Cost	Total_Cost	Currency	No	Calculated from fuel and additional expenses	Total trip expenditure
Fuel Efficiency	Fuel_Efficiency	Decimal	No	Distance divided by fuel used; 2 decimal places	Calculated fuel efficiency in km/L
Cost Per KM	Cost_Per_KM	Decimal	No	Total cost divided by distance; 2 decimal places	Average cost per kilometre

FUEL_DETAILS SUBFORM					
Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Fuel Date	Fuel_Date	Date	Yes	Required	Date of fuel purchase
Litres	Litres	Decimal	Yes	Positive numeric value	Quantity of fuel purchased
Amount	Amount	Currency	Yes	Required	Fuel purchase amount
Fuel Station	Fuel_Station	Single Line	No	Optional	Name of the fuel station

ADDITIONAL_EXPENSES SUBFORM					
Field Label	**Link Name**	**Field Type**	**Mandatory**	**Validation / Default**	**Remarks**
Expense Type	Expense_Type	Drop Down	Yes	Toll / Parking / Loading / Other	Type of additional trip expense
Amount	Amount	Currency	Yes	Required	Expense amount
Description	Description	Multi Line	No	Optional	Additional expense details

4.4 Process and Workflow Design

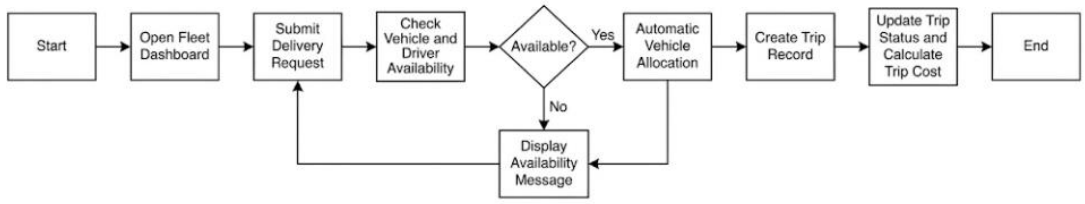


Figure 4.3: Fleet Operations Process Flow

4.5 User Interface and Navigation Design

The user interface of the Fleet Operations and Vehicle Allocation System is organised around the main fleet management activities. The Fleet_Dashboard page acts as the central page for viewing important operational information.
The dashboard contains metric cards showing the total number of vehicles, available vehicles, pending deliveries and completed trips. It also contains a pie chart showing vehicle availability and another pie chart showing delivery request status.
The application provides separate forms for maintaining vehicle, driver, delivery request and trip records. Reports such as All_Vehicles, All_Drivers, Pending_Requests and Trips allow users to access the corresponding information.

The navigation structure separates data entry and record monitoring components, allowing users to move between fleet records, delivery information, trip details and dashboard summaries according to their assigned access permissions.

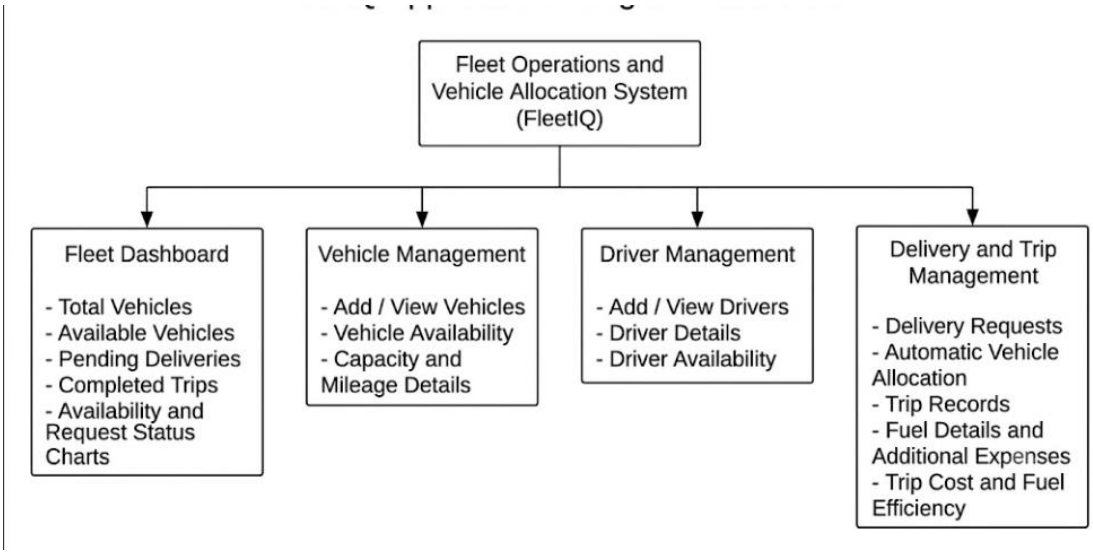


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

The application uses three permission profiles: Admin_Access, Manager_Access and Driver_Access. These profiles define the access rights available to different categories of users.

The Admin_Access profile provides administrative access to the vehicle, driver, delivery request and trip records. It also provides viewing access to the Fleet_Dashboard page.

The Manager_Access profile allows users to view and edit vehicle, driver, delivery request and trip records. Delete access is not provided for these records. Managers can also view the Fleet_Dashboard page.

The Driver_Access profile provides access to the Trips form with view and edit permissions. It does not provide access to the Vehicles, Drivers or Delivery_Requests forms. Drivers can view the Fleet_Dashboard page.

The permission structure helps organise application access according to the responsibilities of each role.

Table 4.4: Role–Permission Matrix

Component	Administrator	Standard User	Portal User	Notes
Form: Vehicles	A E V D	A E V	—	Administrator uses Admin_Access; Standard User uses Manager_Access.
Form: Drivers	A E V D	A E V	—	Driver_Access does not have access to the Drivers form.
Form: Delivery_Requests	A E V D	A E V	—	Administrator and Manager can manage delivery request records.

Component	Administrator	Standard User	Portal User	Notes
Form: Trips	A E V D	A E V	A E V	Driver_Access has View and Edit permissions for Trips; access is governed by the configured profile permissions.
Report: All_Vehicles	V	V	—	Vehicle report is available to Administrator and Manager.
Report: All_Drivers	V	V	—	Driver report is available to Administrator and Manager.
Report: Pending_Requests	V	V	—	Pending delivery requests can be viewed by Administrator and Manager.
Report: Trips	V	V	V	Trip report access follows the corresponding Trips permissions.
Page: Fleet_Dashboard	V	V	V	Dashboard is viewable by all three configured profiles.

Permission notation:
A = Add
E = Edit
V = View
D = Delete
— = No access

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Fleet Operations and Vehicle Allocation System was developed using Zoho Creator Trial Edition as a low-code application. The application was designed to support the management of vehicles, drivers, delivery requests and trips through a centralised interface.

The application contains separate sections for Vehicles, Drivers, Delivery_Requests and Trips. These sections allow authorised users to enter, view and maintain fleet-related information. The Fleet_Dashboard page provides an overview of the fleet operations through summary cards and charts.

The application uses Zoho Creator forms to capture information, reports to display stored records and Deluge workflows to automate selected operations. The application logic includes automatic vehicle and driver allocation, trip cost calculation, fuel efficiency calculation and cost-per-kilometre calculation.

The application was configured with three permission profiles: Admin_Access, Manager_Access and Driver_Access. These profiles organise access to the application components according to the responsibilities assigned to each user.

The completed application was published through the Zoho Creator environment.

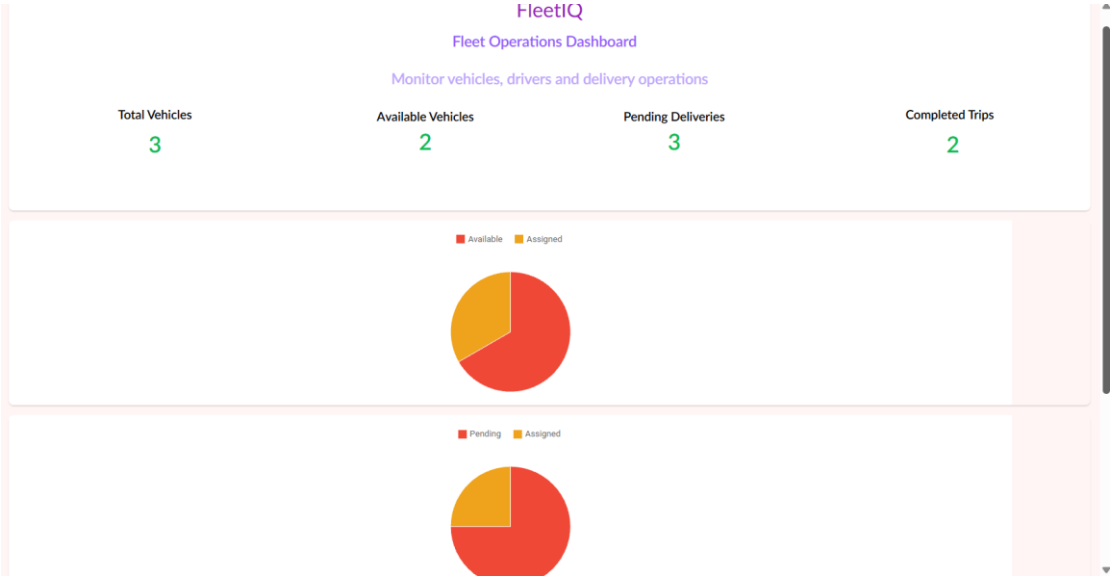


Figure 5.1: Fleet Operations Application Dashboard and Navigation Sections

5.2 Forms and Data Capture

The application consists of four primary forms: Vehicles, Drivers, Delivery_Requests and Trips. Each form captures information required for fleet operations.

The Vehicles form stores Vehicle_ID, Vehicle_Name, Vehicle_Type, Capacity_KG, Mileage_KM_L and Availability. The Vehicle_Type field provides the options Van, Mini Truck and Truck. The Availability field indicates whether a vehicle is Available, Assigned or under Maintenance.

The Drivers form captures Driver_Name, Phone_Number, License_Number and Availability. The availability status is used by the allocation workflow to identify drivers who can be assigned to delivery requests.

The Delivery_Requests form records Request_ID, Source, Destination, Load_Weight_KG, Priority and Request_Status. The Priority field classifies requests as High, Medium or Low, while Request_Status indicates whether a request is Pending, Assigned or Completed.

The Trips form connects a delivery request with its allocated vehicle and driver through lookup fields. It also records Distance_KM, Fuel_Used_L and Trip_Status. Additional calculated fields include Total_Cost, Fuel_Efficiency and Cost_Per_KM.

Two subforms are included in the Trips form. Fuel_Details records fuel purchase information, including date, litres, amount and fuel station. Additional_Expenses records other trip-related expenses such as toll, parking, loading and other charges.

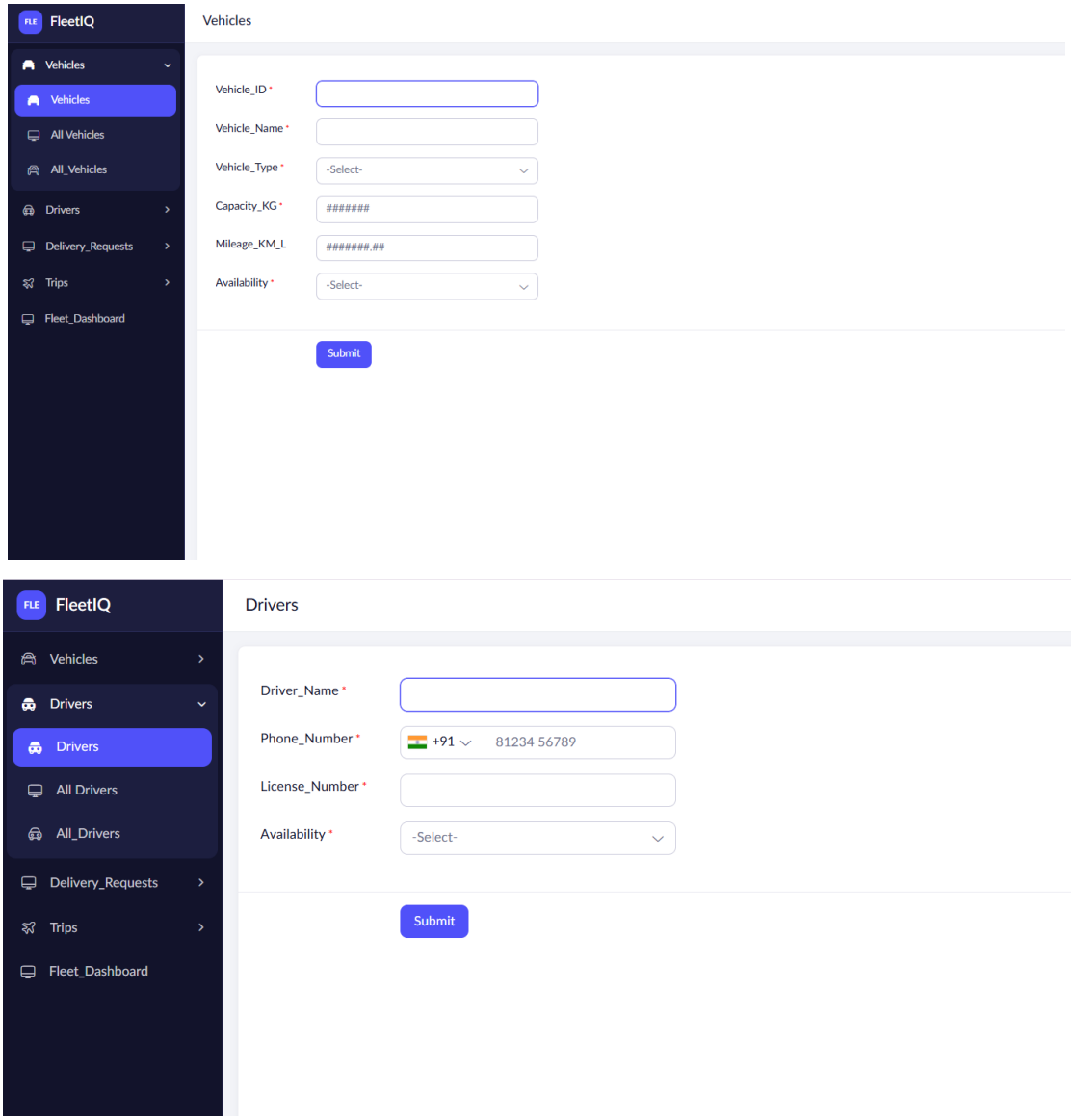


Figure 5.2: Vehicles and Drivers Data Forms

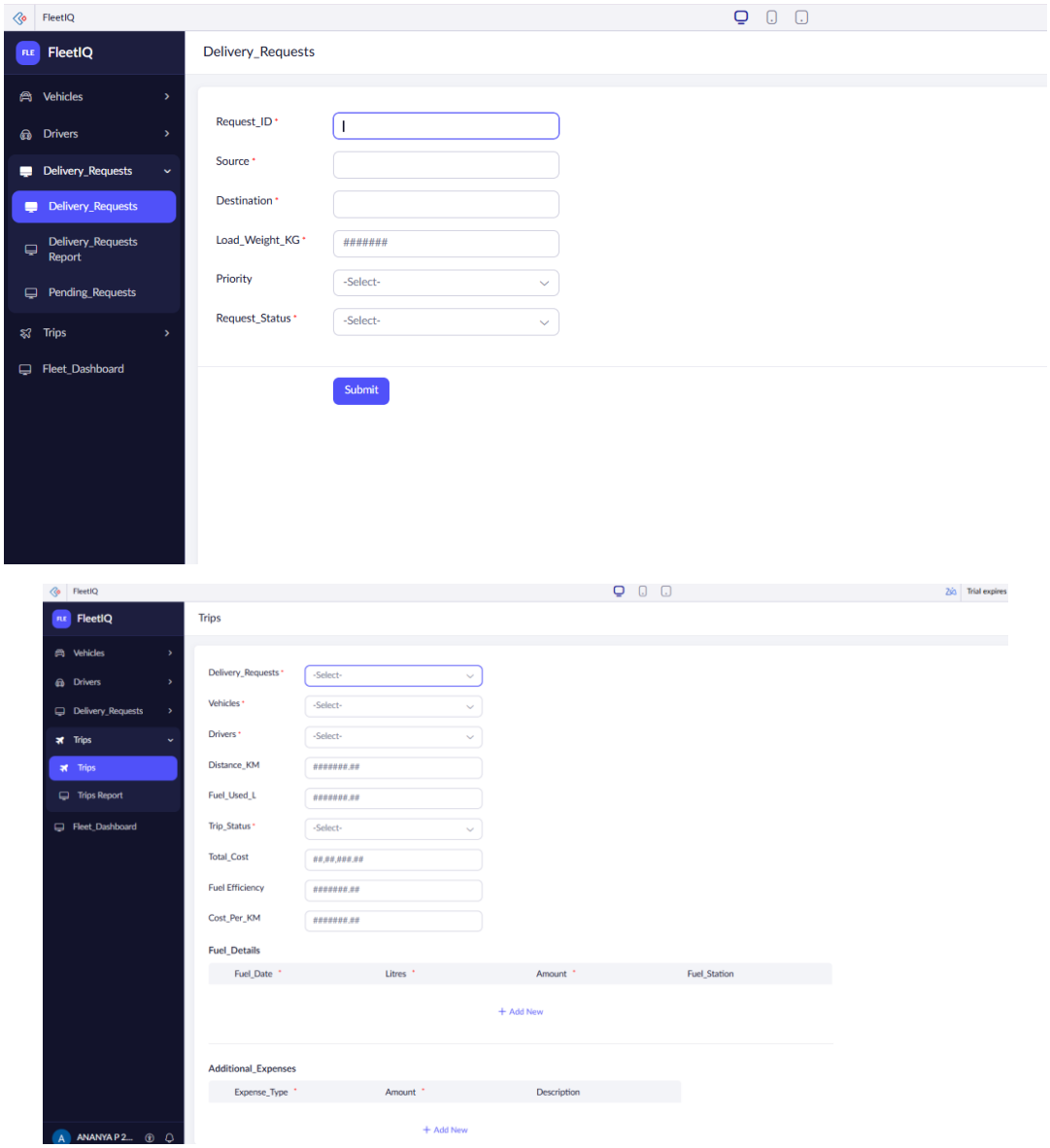


Figure 5.3: Delivery Request and Trip Forms

5.3 Reports

The application uses Zoho Creator reports to display stored fleet information. Reports provide a structured view of vehicle, driver, delivery request and trip records.

The All_Vehicles report displays the registered vehicle details, including vehicle type, capacity, mileage and availability. The All_Drivers report displays driver information and availability status.

The Pending_Requests report displays delivery requests whose Request_Status is Pending. This allows authorised users to identify requests that require allocation.

The Trips report displays trip records along with their associated delivery requests, vehicles, drivers, trip status and calculated values. These reports help users review fleet records without opening each individual form.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All_Vehicles	List report	Vehicles	Displays vehicle records and availability information	Standard report controls	Administrator, Manager
All_Drivers	List report	Drivers	Displays driver details and availability information	Standard report controls	Administrator, Manager
Pending_Requests	List report	Delivery_Requests	Filtered by Request_Status = Pending	Standard report controls	Administrator, Manager
Trips	List report	Trips	Displays trip records, linked resources and calculated trip values	Standard report controls	Administrator, Manager, Driver (according to configured access)
Fleet Dashboard Summary	Dashboard / KPI cards	Vehicles, Delivery_Requests and Trips	Displays fleet counts and operational summaries	Dashboard components	Administrator, Manager, Driver

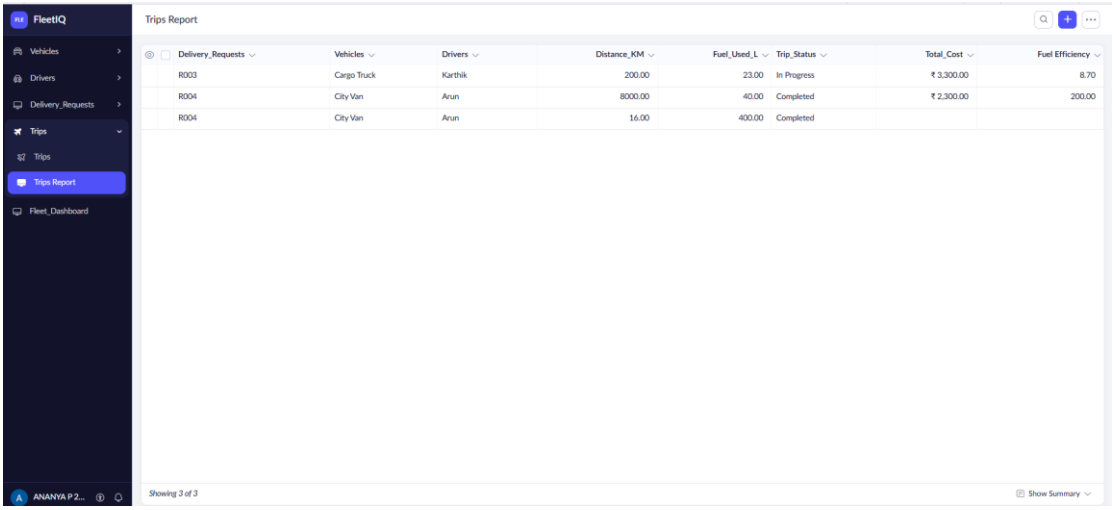


Figure 5.4: Fleet Operations Reports

5.4 Pages and Dashboards

The Fleet_Dashboard page provides a consolidated view of important fleet operation information. It contains four summary cards that display Total Vehicles, Available Vehicles, Pending Deliveries and Completed Trips.

The Total Vehicles card displays the total number of vehicle records stored in the Vehicles form. The Available Vehicles card displays the number of vehicles whose availability status is Available. The Pending Deliveries card displays the number of delivery requests whose status is Pending. The Completed Trips card displays the number of trip records whose status is Completed.

The dashboard also contains two pie charts. The first chart represents vehicle availability by displaying the distribution of vehicles across their availability statuses. The second chart represents delivery request status by displaying the distribution of Pending, Assigned and Completed requests.

A Delivery_Requests report component is also embedded in the dashboard page to provide direct access to delivery request information.

The dashboard helps users monitor fleet resources and delivery progress through a single interface.

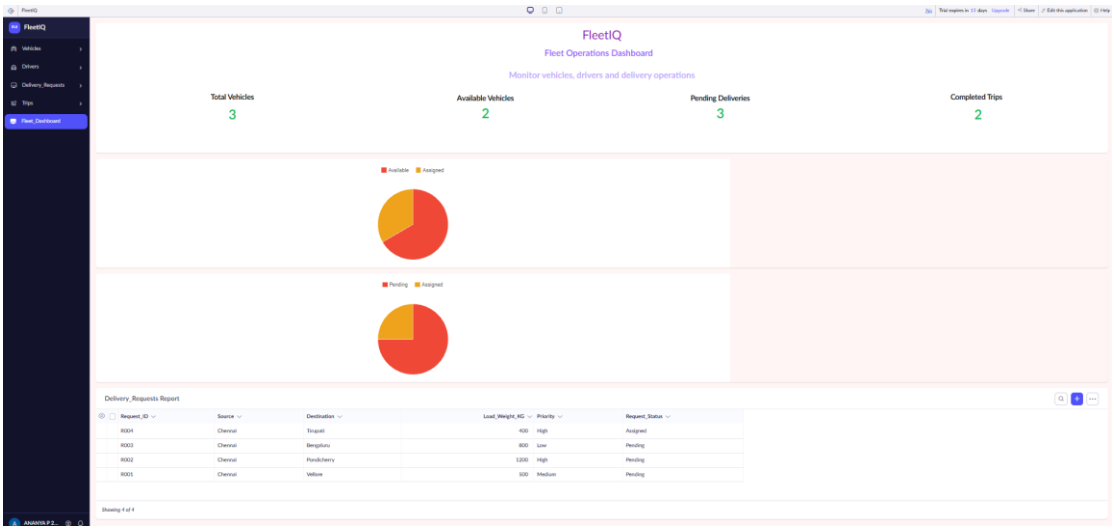


Figure 5.5: Fleet Operations Dashboard

5.5 Deluge Scripting

Deluge scripting is used to automate allocation and trip-related calculations in the application. The scripts are associated with Zoho Creator form workflows and execute when the configured form events occur.

The main allocation workflow is named Auto_Vehicle_Allocation. It is associated with the Delivery_Requests form and runs after successful form submission. It checks vehicle availability and capacity against the requested load weight. It also checks driver availability. When suitable resources are found, the workflow creates a trip record and updates the corresponding resource and request statuses.

The trip cost calculation workflow calculates the total trip expenditure by adding the amounts recorded in the Fuel_Details and Additional_Expenses subforms.

The Fuel Efficiency Calculation workflow calculates fuel efficiency using the distance travelled and fuel consumed. The Fuel Efficiency Update workflow performs the corresponding calculation when the trip record is edited.

The Cost Per KM Calculation workflow calculates the average trip cost per kilometre using the total trip cost and distance travelled. A corresponding update workflow recalculates the value when the trip record is edited.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Delivery_Requests – Created; after successful form submission	Identifies an available vehicle with sufficient capacity and an available driver, creates a trip record and updates allocation statuses	Form input, record fetch, conditional logic, criteria-based search, record insertion and record update
DS-02	Trips – Created or Edited; after successful form submission	Calculates the total trip cost using Fuel_Details and Additional_Expenses	Subform access, iteration, conditional logic, arithmetic and record update
DS-03	Trips – Created; after successful form submission	Calculates fuel efficiency using distance travelled and fuel consumed	Form input, conditional logic, arithmetic and record update
DS-04	Trips – Edited; after successful form submission	Recalculates fuel efficiency when trip information is updated	Form input, conditional logic, arithmetic and record update
DS-05	Trips – Created or Edited; after successful form submission	Calculates cost per kilometre using total trip cost and distance travelled	Record fetch, conditional logic, arithmetic and record update

The Auto_Vehicle_Allocation workflow is triggered when a delivery request is successfully submitted. It retrieves vehicles whose Availability is Available and whose

Capacity_KG is greater than or equal to the requested Load_Weight_KG. It also retrieves drivers whose Availability is Available.

If both an eligible vehicle and an available driver are found, the workflow creates a record in the Trips form. The selected vehicle and driver are then marked as Assigned, and the delivery request status is updated to Assigned.

The workflow was tested using delivery request R004, with a load weight of 400 kg and High priority. The request was successfully allocated, and a corresponding trip record was created.

The trip cost calculation workflow adds the amounts entered in the Fuel_Details and Additional_Expenses subforms. For the tested trip, fuel expenses of ₹2,000 and additional expenses of ₹300 produced a total trip cost of ₹2,300.

The fuel efficiency calculation was tested using a distance of 200 km and fuel consumption of 20 litres. The resulting fuel efficiency was 10 km/L.

The cost-per-kilometre calculation was tested using a total trip cost of ₹2,300 and a distance of 200 km. The resulting cost per kilometre was ₹11.50.

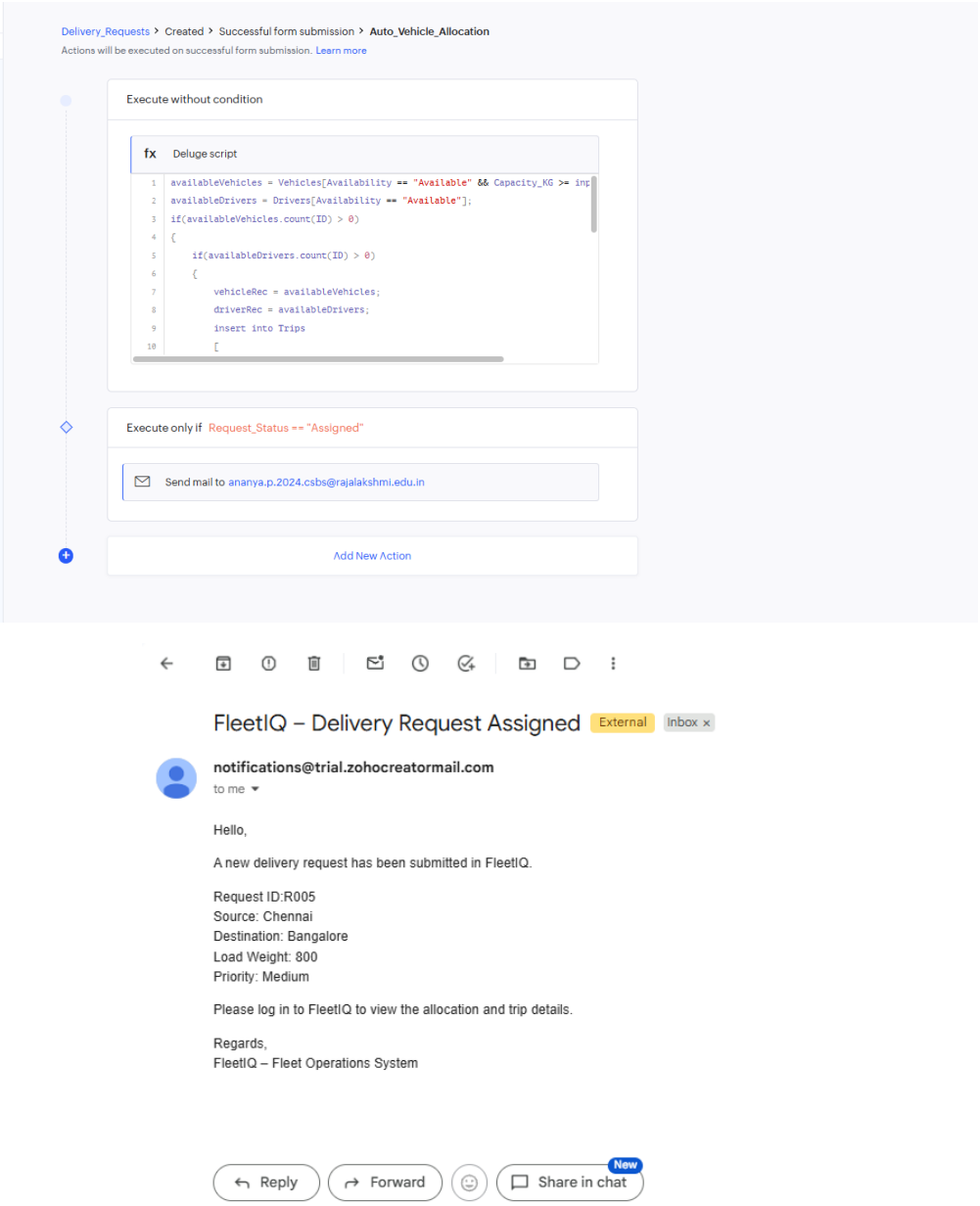


Figure 5.6: Delivery Request Workflow and Deluge Resource Allocation

Code 5.1: Auto_Vehicle_Allocation – Deluge Script of the Form Workflow (DS-01)

```
// Script ID : DS-01
// Workflow  : Auto_Vehicle_Allocation
// Trigger   : Delivery_Requests -> Created -> Successful form submission

availableVehicles = Vehicles[Availability == "Available" && Capacity_KG >=
input.Load_Weight_KG];
availableDrivers = Drivers[Availability == "Available"];

if(availableVehicles.count(ID) > 0)
{
    if(availableDrivers.count(ID) > 0)
    {
        vehicleRec = availableVehicles;
        driverRec = availableDrivers;

        insert into Trips
        [
            Delivery_Requests = input.ID
            Vehicles = vehicleRec.ID
            Drivers = driverRec.ID
            Trip_Status = "Assigned"
        ];

        vehicleRec.Availability = "Assigned";
        driverRec.Availability = "Assigned";
        input.Request_Status = "Assigned";
    }
    else
    {
        info "No available driver found.";
    }
}
else
{
    info "No suitable vehicle found.";
}
```

How the script works:

1. **Identify suitable vehicles:** The script searches the Vehicles form for records whose Availability is Available and whose Capacity_KG is sufficient for the requested load weight.
2. **Check driver availability:** It retrieves driver records whose Availability is marked as Available.
3. **Verify resource availability:** The count() function checks whether at least one suitable vehicle and one available driver exist.
4. **Create the trip:** When both resources are available, a new Trips record is created with lookup references to the delivery request, vehicle and driver.

5. **Update statuses:** The selected vehicle and driver are marked as Assigned, and the delivery request status is changed to Assigned.
6. **Handle unsuccessful allocation:** If no suitable vehicle or available driver is found, the workflow displays an appropriate information message.

Code 5.2: Fuel Efficiency Calculation – Deluge Script (DS-03)

```
// Script ID : DS-03
// Purpose   : Calculate fuel efficiency in km/L

if(input.Fuel_Used_L != null && input.Fuel_Used_L > 0)
{
    tripRecord = Trips[ID == input.ID];
    tripRecord.Fuel_Efficiency = input.Distance_KM / input.Fuel_Used_L;
}
else
{
    info "Enter valid fuel usage to calculate fuel efficiency.";
}
```

How the script works:

1. **Check fuel consumption:** The script verifies that Fuel_Used_L is not empty and is greater than zero.
2. **Retrieve the trip:** The corresponding trip record is fetched using its ID.
3. **Calculate efficiency:** Distance_KM is divided by Fuel_Used_L to calculate fuel efficiency.
4. **Update the record:** The calculated value is stored in Fuel_Efficiency.

Code 5.3: Cost Per KM Calculation – Deluge Script (DS-05)

```
// Script ID : DS-05
// Purpose   : Calculate cost per kilometre

tripRecord = Trips[ID == input.ID];

if(input.Distance_KM != null && input.Distance_KM > 0)
{
    tripRecord.Cost_Per_KM = tripRecord.Total_Cost / input.Distance_KM;
}
```

How the script works:

1. Retrieve the trip record using its ID.
2. Check that the distance is available and greater than zero.
3. Divide Total_Cost by Distance_KM.

4. Store the calculated value in Cost_Per_KM.

Note: The complete final script for total trip cost should be copied directly from the saved Zoho Creator workflow into the report. The workflow calculates the sum of Fuel_Details.Amount and Additional_Expenses.Amount.

5.6 Workflows and Automation

The application uses Zoho Creator form workflows to automate resource allocation and trip-related calculations. These workflows reduce the need for users to perform repeated calculations manually.

The Auto_Vehicle_Allocation workflow is associated with the Delivery_Requests form. It runs after a new delivery request is successfully submitted. The workflow checks the capacity and availability of vehicles and the availability of drivers. When suitable resources are found, it creates a trip record and updates the corresponding statuses.

The trip cost calculation workflow runs when a Trips record is created or edited. It calculates the total expenditure by adding fuel expenses and additional expenses.

The fuel efficiency workflows calculate the distance travelled per litre of fuel consumed. The cost-per-kilometre workflows calculate the total trip expenditure divided by the distance travelled.

The dashboard components display information from the stored records and provide an overview of the current fleet operations.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Auto_Vehicle_Allocation	Form workflow	Created – after successful form submission	Delivery request submitted	Find suitable vehicle and available driver; create Trips record; update vehicle, driver and request statuses	DS-01
Trip Cost Calculation	Form workflow	Created or Edited – after successful form submission	Trip record submitted or updated	Sum Fuel_Details and Additional_Expenses amounts and update Total_Cost	DS-02

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Fuel Efficiency Calculation	Form workflow	Created – after successful form submission	Fuel_Used_L is greater than zero	Calculate distance divided by fuel consumed and update Fuel_Efficiency	DS-03
Fuel Efficiency Update	Form workflow	Edited – after successful form submission	Fuel usage and distance are valid	Recalculate Fuel_Efficiency	DS-04
Cost Per KM Calculation	Form workflow	Created or Edited – after successful form submission	Distance_KM is greater than zero	Calculate Total_Cost divided by Distance_KM and update Cost_Per_KM	DS-05
Dashboard Summary	Report / page display	Open Fleet_Dashboard	Existing records available	Display fleet summary cards, charts and delivery request report	Built-in display components

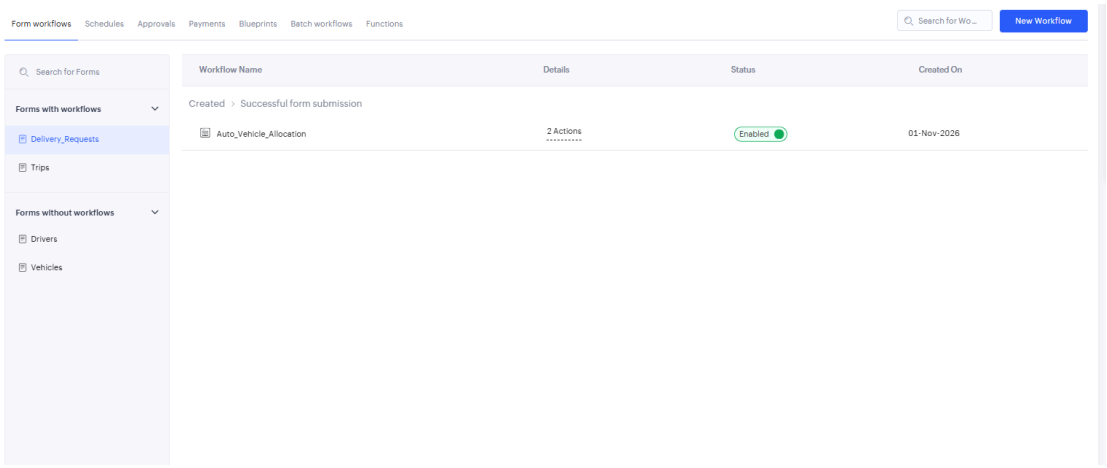


Figure 5.7: Zoho Creator Workflow Configuration for Fleet Operations

5.7 Integrations

The Fleet Operations and Vehicle Allocation System is implemented using Zoho Creator's built-in application components. The forms, lookup fields, reports, dashboard pages and Deluge workflows operate within the Zoho Creator environment.

The Trips form uses lookup fields to connect delivery requests, vehicles and drivers. These relationships allow trip records to reference the corresponding records in the other forms.

The application does not depend on external REST APIs, webhooks or third-party integration services. The required allocation and calculation operations are performed using Zoho Creator workflows and Deluge scripting.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Internal form relationship	Zoho Creator Forms	Lookup fields	Platform-managed	Connects Trips with Delivery_Requests, Vehicles and Drivers.
Workflow automation	Zoho Creator Deluge	Form workflows	Platform-managed	Performs automatic resource allocation and trip-related calculations.
Dashboard and reports	Zoho Creator Pages and Reports	Built-in components	Platform-managed	Displays stored fleet information, summary cards and charts.

No external API, REST call, webhook or third-party service is used in the current implementation. The application uses Zoho Creator's built-in features for data management and automation.

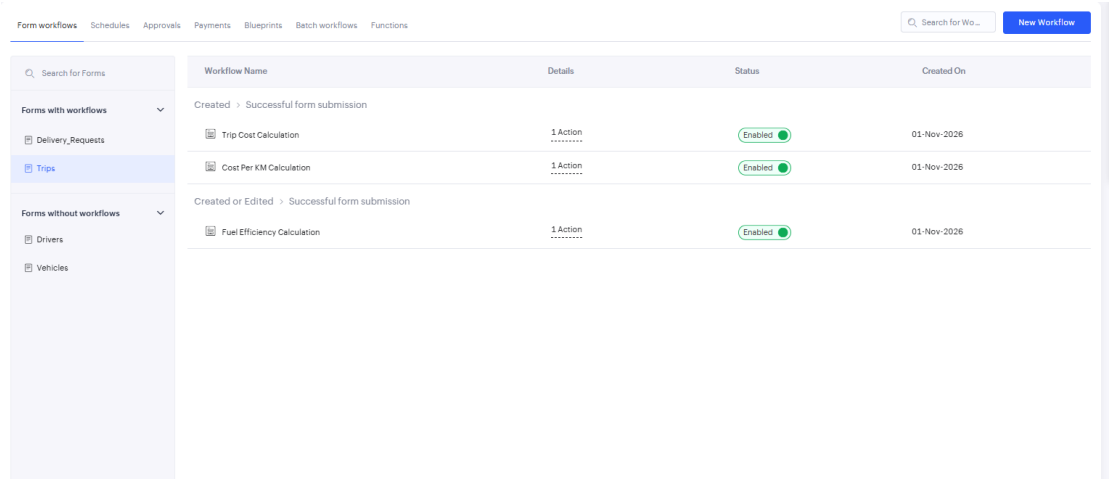


Figure 5.8: Workflow Module of Zoho Creator

5.8 User Management and Role-Based Access

The application uses three permission profiles to organise access to its forms, reports and dashboard page. The profiles are Admin_Access, Manager_Access and Driver_Access.

The Admin_Access profile provides administrative access to the Vehicles, Drivers, Delivery_Requests and Trips forms. It allows the administrator to view, add, edit and delete records. The Fleet_Dashboard page is available for viewing.

The Manager_Access profile allows managers to view and edit vehicle, driver, delivery request and trip records. Delete permissions are not provided for these forms. Managers can also view the Fleet_Dashboard page.

The Driver_Access profile provides view and edit permissions for the Trips form. Access to the Vehicles, Drivers and Delivery_Requests forms is not provided through this profile. Drivers can also view the Fleet_Dashboard page.

These profiles organise access according to the responsibilities of each application role and help prevent users from accessing components that are not required for their tasks.

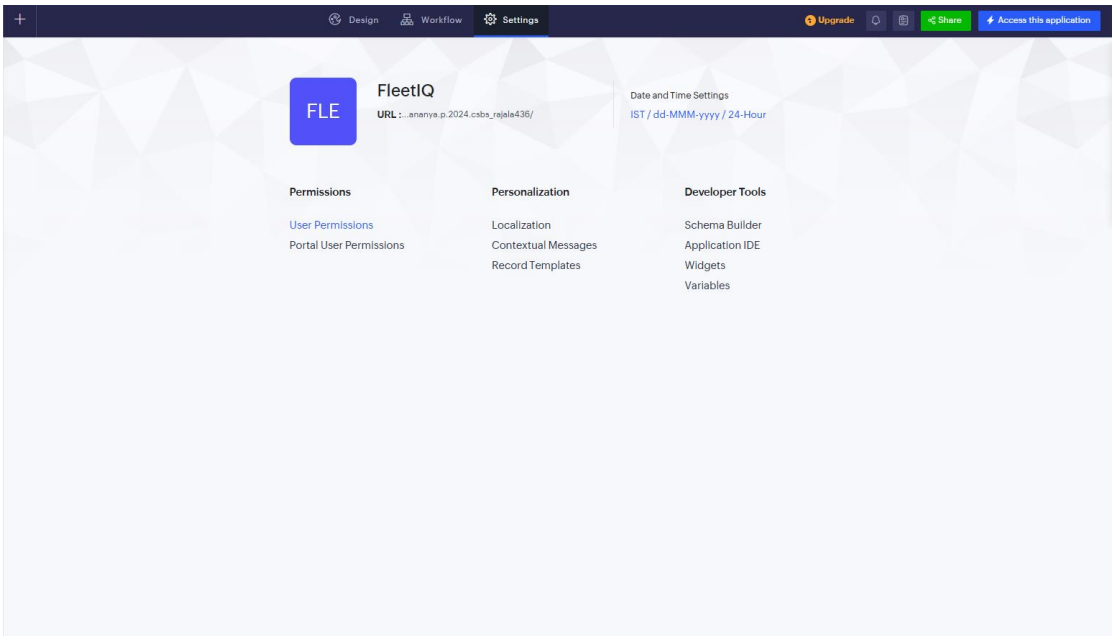


Figure 5.9: Application Settings Showing User Permission Profiles

5.9 AI and Analytics Features

The Fleet Operations and Vehicle Allocation System provides operational summaries through the Fleet_Dashboard page. The dashboard contains four metric cards: Total Vehicles, Available Vehicles, Pending Deliveries and Completed Trips.

The vehicle availability pie chart displays the distribution of vehicles according to their current availability status. The delivery request status pie chart displays the distribution of requests according to their Pending, Assigned and Completed statuses.

The dashboard uses the stored records in the Vehicles, Delivery_Requests and Trips forms to display these summaries. These components help users understand the current state of fleet resources and delivery operations.

The application also performs rule-based resource allocation through Deluge scripting. It checks vehicle capacity and availability along with driver availability before creating a trip record. This is a predefined rule-based workflow and does not use artificial intelligence or machine learning.

No predictive model, AI-based allocation, OCR or external Zoho Analytics integration is used in the current application. The dashboard is therefore described as a visual reporting and monitoring feature built using Zoho Creator components.

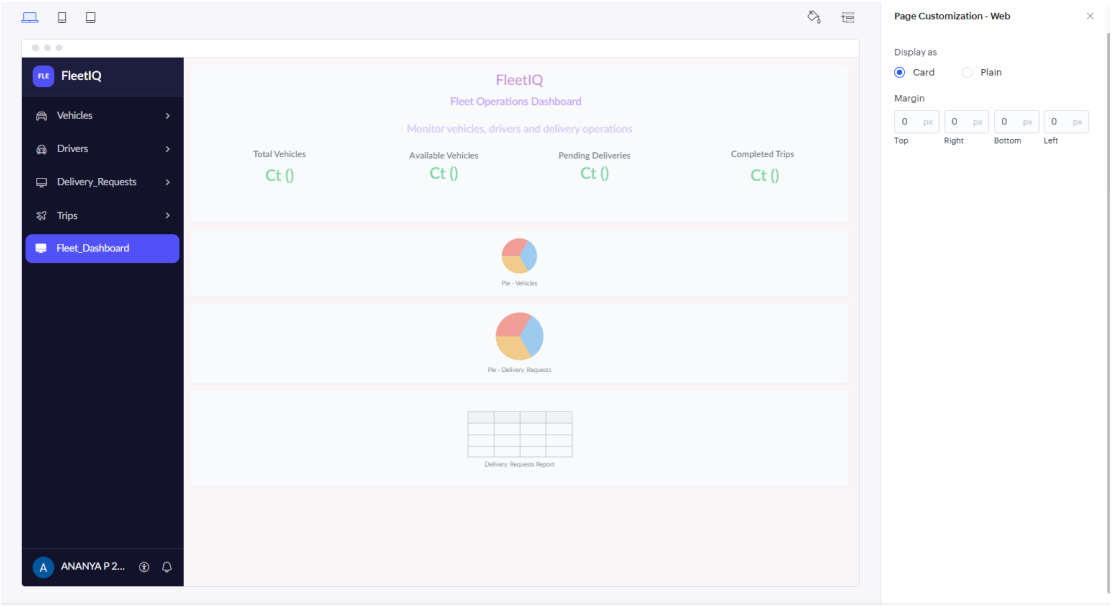


Figure 5.10: Fleet Dashboard in the Page Builder Showing Summary Cards and Chart Components

5.10 Pro-Code Extension

The current project is implemented using Zoho Creator forms, reports, lookup fields, pages and Deluge workflows. Deluge scripting is used for resource allocation and trip-related calculations.

No separate pro-code extension such as Zoho Catalyst, custom JavaScript widgets or external REST API services is used in the application. The built-in low-code capabilities of Zoho Creator are sufficient for the current project requirements.

A pro-code extension could be considered in the future if the application requires advanced route optimisation, integration with external fleet management services or more complex data processing. Such extensions are outside the scope of the present implementation.

5.11 Mobile and Portal Access

The application is developed and published through Zoho Creator. Its forms, reports and pages are accessed through the platform's application interface.

Zoho Creator provides mobile access options through its mobile application, allowing users to access supported application components from compatible devices. The actual appearance and availability of individual components may depend on the device and application configuration.

The current implementation focuses on the web-based application and its configured permission profiles. No separate customer portal or external driver portal is included in the project.

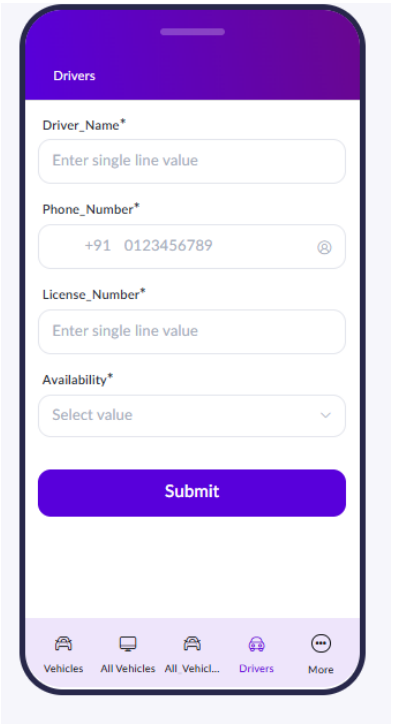


Figure 5.11: Illustrative Mobile Layout of the Fleet Operations Application

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out to verify the major functions of the Fleet Operations and Vehicle Allocation System. The testing process covered form data entry, record creation, automatic resource allocation, trip calculations, dashboard summaries, reports and permission settings.

Form testing involved entering vehicle, driver and delivery request information and verifying that the records were saved correctly. Allocation testing was performed by submitting a delivery request and checking whether the system identified an available vehicle with sufficient capacity and an available driver.

Calculation testing covered total trip cost, fuel efficiency and cost per kilometre. The calculated values were compared with manually derived results to verify the accuracy of the workflows.

Dashboard testing involved checking the four summary cards and two pie charts against the records stored in the application. Report testing involved opening the vehicle, driver, pending request and trip reports.

Permission testing covered the configured Admin_Access, Manager_Access and Driver_Access profiles to ensure that access to the application components followed the intended role structure.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Open Vehicles form	Open the Vehicles form	The vehicle form should be displayed	The form is available in the application	Pass
TC-02	FR-02	Open Drivers form	Open the Drivers form	The driver form should be displayed	The form is available in the application	Pass
TC-03	FR-03	Open Delivery_Requests form	Open the Delivery_Requests form	The delivery request form should be displayed	The form is available in the application	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-04	FR-01	Enter vehicle details	Enter vehicle ID, name, type, capacity and availability	Vehicle details should be accepted and saved	Vehicle records are maintained in the application	Pass
TC-05	FR-02	Enter driver details	Enter driver name, phone number, licence number and availability	Driver details should be accepted and saved	Driver records are maintained in the application	Pass
TC-06	FR-03	Create a delivery request	Enter request ID, source, destination, load weight and priority	The delivery request should be saved	The delivery request was created successfully	Pass
TC-07	FR-04	Check vehicle capacity	Submit a request with a load weight of 400 kg	The system should identify an available vehicle with sufficient capacity	A suitable vehicle was identified during the R004 allocation test	Pass
TC-08	FR-05	Check driver availability	Submit a request when an available driver exists	The system should identify an available driver	An available driver was assigned during the R004 allocation test	Pass
TC-09	FR-05	Verify automatic allocation	Submit delivery request R004 with a load weight of 400 kg and High priority	A trip record should be created with an eligible vehicle and driver	R004 was automatically assigned and a Trips record was created	Pass
TC-10	FR-05	Verify resource status update	Complete the automatic allocation process	The selected vehicle and driver should be marked as Assigned and the request status should be updated	The allocation workflow updates the corresponding statuses	Pass
TC-11	FR-06	Verify total trip cost	Enter fuel expense of ₹2,000 and additional expense of ₹300	Total_Cost should be ₹2,300	Total_Cost was calculated as ₹2,300	Pass
TC-12	FR-07	Verify fuel efficiency	Enter Distance_KM = 200 and Fuel_Used_L = 20	Fuel_Efficiency should be 10 km/L	The calculated fuel efficiency was 10 km/L	Pass
TC-13	FR-07	Verify cost per kilometre	Enter Total_Cost = ₹2,300 and Distance_KM = 200	Cost_Per_KM should be ₹11.50	Calculated cost per kilometre was ₹11.50	Pass
TC-14	FR-08	Open Fleet Dashboard	Open Fleet_Dashboard	The dashboard should display its four metric cards and two charts	The cards and charts were displayed correctly	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-15	FR-08	Verify Total Vehicles card	Compare the card with the vehicle records	The card should display the total number of vehicles	The dashboard card displays the vehicle count	Pass
TC-16	FR-08	Verify Available Vehicles card	Compare the card with vehicle availability statuses	The card should display the number of available vehicles	The dashboard card displays the available vehicle count	Pass
TC-17	FR-08	Verify Pending Deliveries card	Compare the card with pending delivery requests	The card should display the number of pending requests	The dashboard card displays the pending request count	Pass
TC-18	FR-08	Verify Completed Trips card	Compare the card with completed trip records	The card should display the number of completed trips	The dashboard card displays the completed trip count	Pass
TC-19	FR-08	Open vehicle and driver reports	Open All_Vehicles and All_Drivers	The reports should display the corresponding records	The reports are available in the application	Pass
TC-20	FR-08	Open Pending_Requests report	Open the Pending_Requests report	The report should display requests with Pending status	The filtered report is available in the application	Pass
TC-21	NFR-02	Verify role-based access	Review the configured permission profiles	Each profile should provide the configured access to application components	Admin_Access, Manager_Access and Driver_Access profiles are configured	Pass

6.3 Debugging Deluge Scripts

Debugging was performed during the configuration and testing of the Deluge workflows. The process involved checking form link names, lookup field references, workflow triggers and the data types used in calculations.

For automatic vehicle allocation, the criteria used to retrieve vehicles and drivers were checked against the availability fields and requested load weight. The creation of the Trips record and the updating of the related statuses were verified through the R004 test request.

For trip cost calculation, the fuel and additional expense amounts were checked to ensure that both subforms contributed to the total expenditure. A test using ₹2,000 in fuel expenses and ₹300 in additional expenses produced a total cost of ₹2,300.

For fuel efficiency and cost-per-kilometre calculations, conditions were included to prevent division by zero. The calculations were tested using known distance, fuel and cost values.

The dashboard was also checked to ensure that its summary cards and charts displayed the corresponding application records.

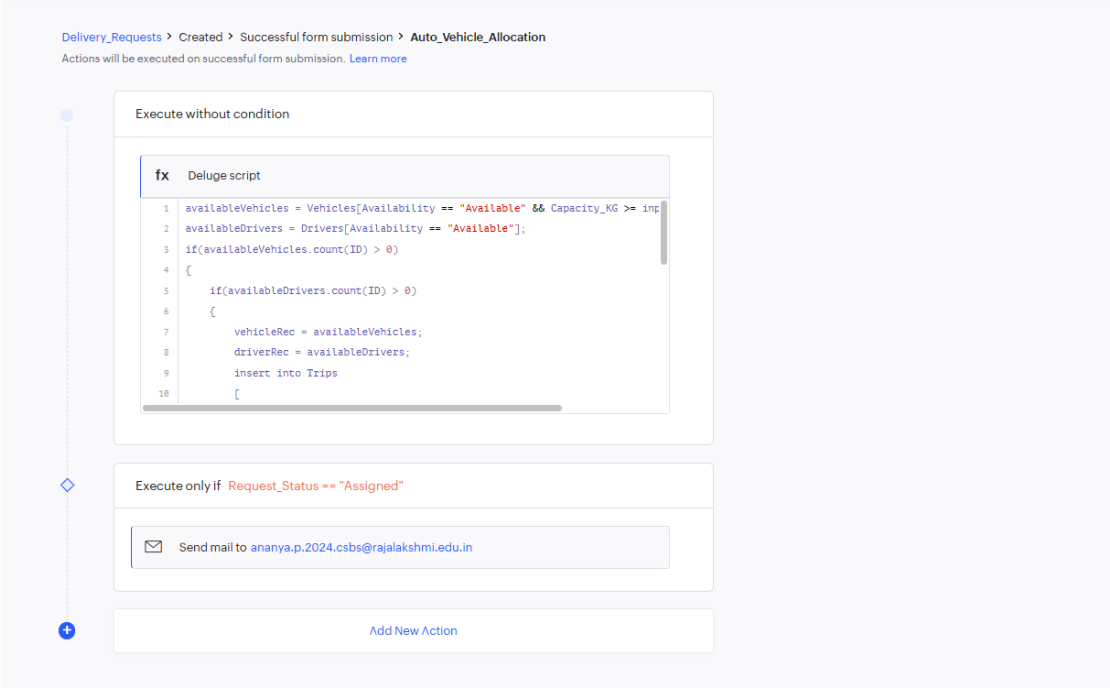


Figure 6.1: Deluge Script Editor Showing the Automatic Vehicle Allocation Workflow

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Automatic allocation must identify vehicles based on load capacity and availability	The allocation process requires both vehicle availability and sufficient load capacity	The Auto_Vehicle_Allocation workflow was configured with criteria to check Availability and Capacity_KG	Closed
BUG-02	Trip total cost must include both fuel and additional expenses	The total trip expenditure requires the amounts from two separate subforms	The trip cost workflow was configured to calculate the combined expense amount	Closed
BUG-03	Fuel efficiency must not be calculated when fuel consumption is zero	Division by zero can occur if fuel usage is zero or invalid	A condition was included to check that Fuel_Used_L is greater than zero	Closed
BUG-04	Cost per kilometre must not be calculated when distance is zero	Division by zero can occur if Distance_KM is zero or invalid	A condition was included to check that Distance_KM is greater than zero	Closed

6.5 User Acceptance Testing

User acceptance testing was carried out by checking the main application functions and confirming that the forms, reports, dashboard and workflows supported the intended fleet management activities.

The application was reviewed for vehicle and driver record management, delivery request creation, automatic allocation, trip record generation and trip-related calculations. The dashboard and reports were also checked to ensure that the stored information was presented in a structured manner.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Application owner	Administrator	The application provides a centralised interface for managing vehicles, drivers, delivery requests and trips	The forms, reports and dashboard were configured to support the required operations.
Application user	Manager	The automatic allocation process reduces the need to manually select a vehicle and driver	The Auto_Vehicle_Allocation workflow was implemented and tested.
Application user	Driver	The Trips section provides access to relevant trip information according to the configured permissions	The Driver_Access profile was configured for the Trips form and dashboard.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The Fleet Operations and Vehicle Allocation System was developed and published through Zoho Creator. The application is accessed through the Zoho Creator environment using an authorised account.

The deployment process involved configuring the application forms, reports, dashboard components, Deluge workflows and permission profiles. The completed application was then published so that its configured components could be accessed through the application interface.

The published application provides the interface for maintaining vehicle records, driver records, delivery requests and trip information. Its dashboard provides a consolidated view of fleet operations.

Before making future modifications, the application owner should review the affected forms and workflows, test the changes, verify the calculated values and ensure that the published application continues to operate correctly. A backup or export should be retained before major changes.

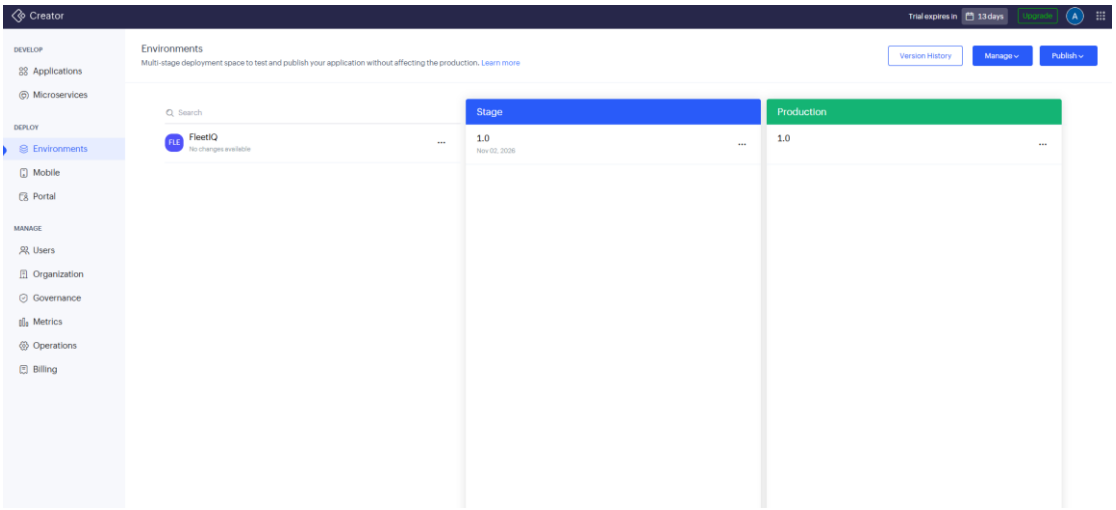


Figure 7.1: Published Fleet Operations Application in Zoho Creator

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	Configure the Vehicles form	Yes
2	Configure the Drivers form	Yes
3	Configure the Delivery_Requests form	Yes
4	Configure the Trips form and subforms	Yes
5	Configure reports for the main application forms	Yes
6	Configure the Fleet_Dashboard page and summary cards	Yes
7	Configure automatic vehicle and driver allocation	Yes
8	Configure trip cost calculation	Yes
9	Configure fuel efficiency calculation	Yes
10	Configure cost-per-kilometre calculation	Yes
11	Configure Admin_Access, Manager_Access and Driver_Access profiles	Yes
12	Test the main application workflows	Yes
13	Publish the completed application	Yes

7.2 Performance Optimisation

The application uses Zoho Creator's built-in forms, reports, lookup fields and dashboard components to organise fleet information. The use of separate forms helps maintain structured records for vehicles, drivers, delivery requests and trips.

The allocation workflow retrieves records using availability and capacity criteria. This avoids requiring the user to manually inspect every vehicle and driver record during the allocation process.

The dashboard uses summary cards and charts to provide a quick overview of fleet operations. Reports provide access to the underlying records when more detailed information is required.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Use criteria-based vehicle selection	Auto_Vehicle_Allocation	Manual identification of suitable vehicles	Workflow retrieves available vehicles with sufficient capacity	Reduces manual resource selection.
Use availability-based driver selection	Auto_Vehicle_Allocation	Manual checking of driver availability	Workflow retrieves available drivers	Supports automatic driver allocation.
Use calculated trip metrics	Trips form	Manual calculation of fuel efficiency and cost per kilometre	Values are calculated through Deluge workflows	Reduces repeated manual calculations.
Use summary cards and charts	Fleet_Dashboard	Reviewing individual records to understand fleet status	Four summary cards and two charts are displayed	Provides a consolidated visual overview.
Use separate reports	All_Vehicles, All_Drivers, Pending_Requests and Trips	Reviewing records through individual forms	Records are available through dedicated reports	Improves record organisation and accessibility.

Note: No formal response-time benchmark was conducted. Improvements are described in terms of workflow and usability rather than unsupported numerical performance claims.

7.3 Security Features

The application uses Zoho Creator account access and configured permission profiles to organise access to its components.

The Admin_Access profile provides administrative permissions for the main forms. The Manager_Access profile allows managers to view and edit operational records without delete permissions. The Driver_Access profile provides access to the Trips form and Fleet_Dashboard page according to the configured permissions.

The application stores its information within the Zoho Creator environment. Form and report access is controlled through the configured application permissions.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho Creator account sign-in	Application access	Reduces unauthorised access to the application.
Permission profiles	Admin_Access, Manager_Access and Driver_Access	Controls access to application forms and pages according to assigned roles.
Form-level permissions	Vehicles, Drivers, Delivery_Requests and Trips	Restricts actions according to the permissions configured for each profile.
Avoid storing credentials in scripts	Deluge workflows	Reduces the risk of exposing sensitive account information through application scripts.

7.4 Governance and Administration

The application should be maintained by its administrator. Recommended practices include keeping form and field names consistent, reviewing vehicle and driver availability, checking delivery request records and verifying trip calculations.

User access should be reviewed whenever responsibilities change. Permission profiles should provide only the access required for each role.

Before modifying a workflow, the administrator should review the affected form fields and test the updated logic with suitable sample records. A backup or export should be retained before major modifications.

Records should be entered accurately so that allocation and calculation workflows operate using valid information. Periodic review of reports and dashboard values can help identify incomplete or inconsistent records.

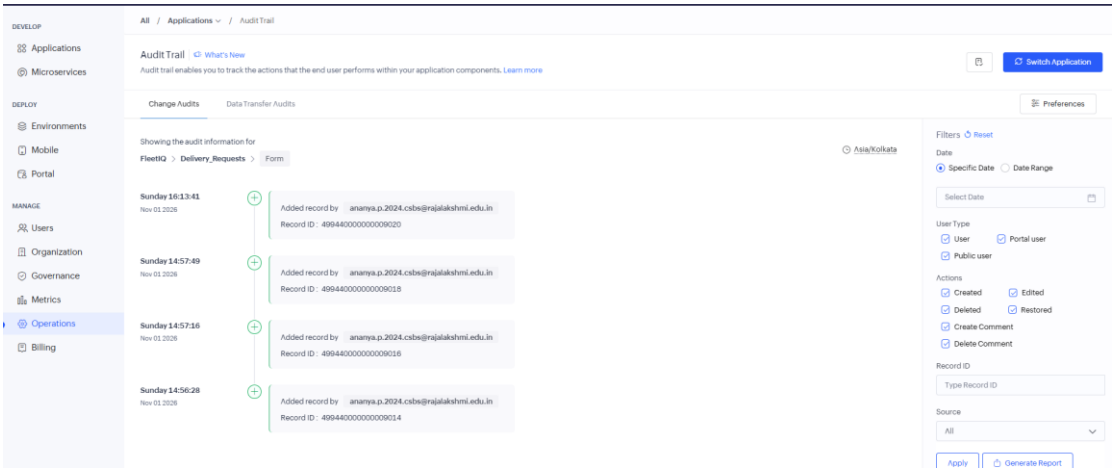


Figure 7.2: Zoho Creator Application Administration and Permission Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A typical usage scenario begins when the user opens the Fleet_Dashboard page to review the current fleet summary. The dashboard displays the total number of vehicles, available vehicles, pending deliveries and completed trips through its four summary cards.

The user can then open the Vehicles and Drivers forms to review or maintain the available fleet resources. When a new delivery requirement is received, the user opens the Delivery_Requests form and enters the request ID, source, destination, load weight and priority.

After the request is submitted, the Auto_Vehicle_Allocation workflow checks vehicle capacity and availability along with driver availability. When suitable resources are found, a trip record is automatically created and the corresponding statuses are updated.

The user can then open the Trips form to review the allocated vehicle and driver and enter trip-related information. Fuel purchases and additional expenses are recorded through the corresponding subforms.

The application calculates total trip cost, fuel efficiency and cost per kilometre using the configured workflows. The resulting information can be viewed through the Trips report.

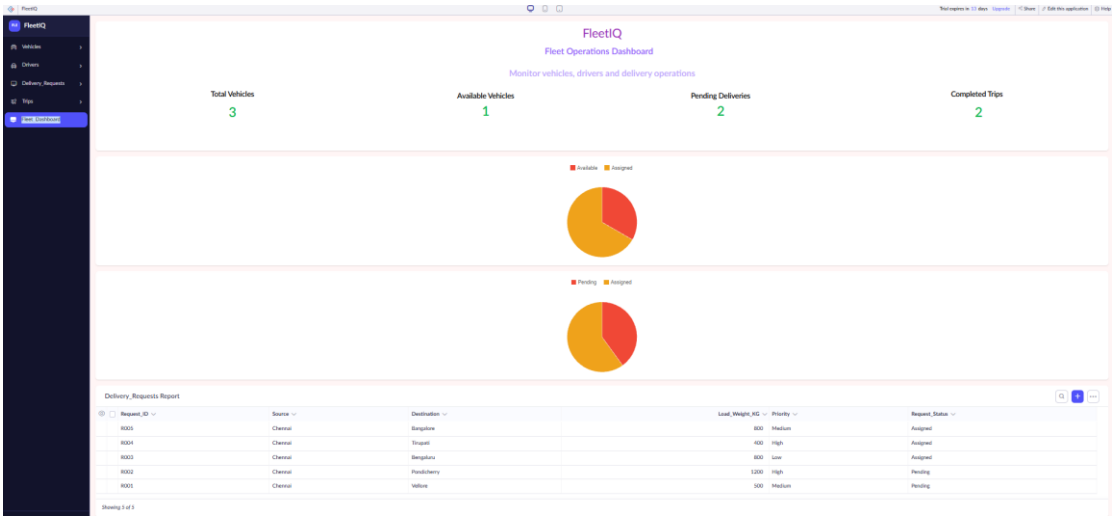


Figure 8.1: Step 1 – Review the Fleet Dashboard

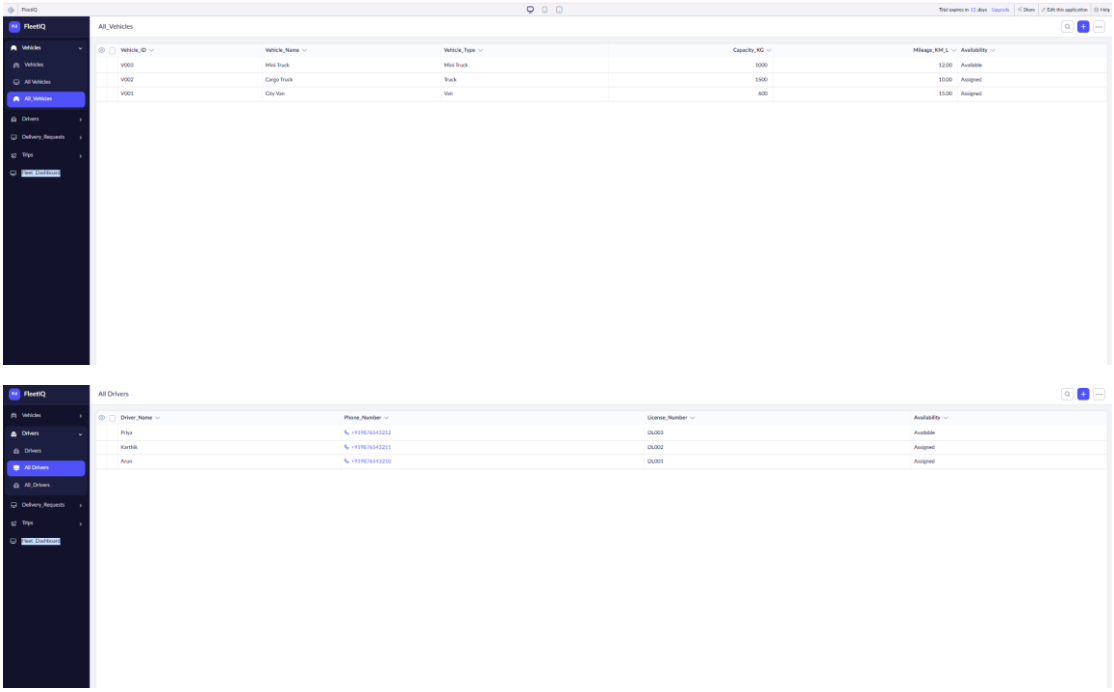


Figure 8.2: Step 2 – Review Vehicle and Driver Records

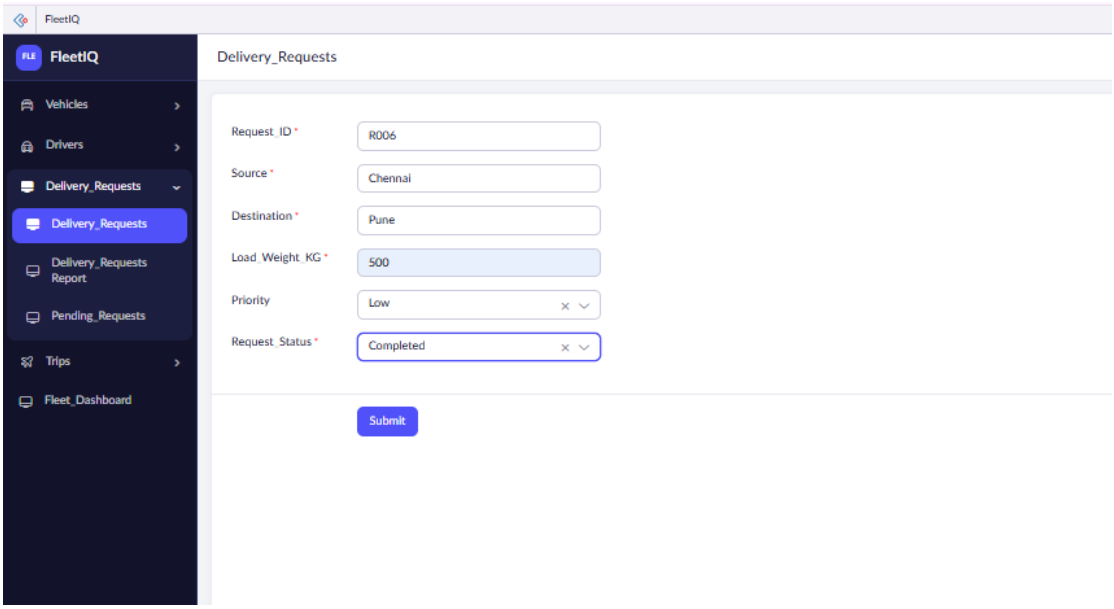


Figure 8.3: Step 3 – Create and Submit a Delivery Request

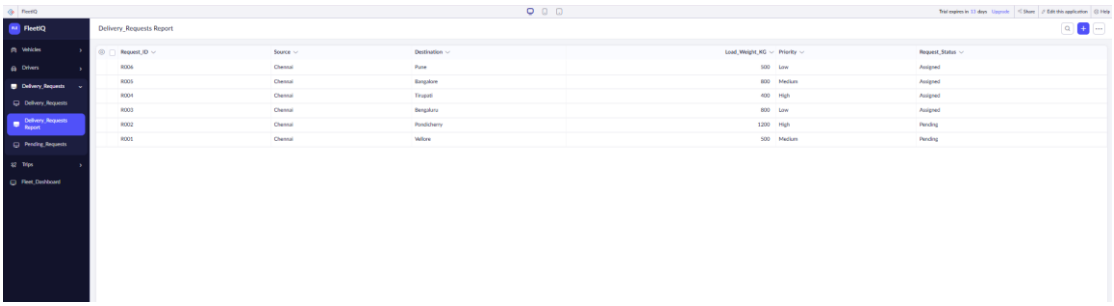


Figure 8.4: Step 4 – View the Automatically Created Trip Record

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Maintain centralised vehicle information	Vehicles form and All_Vehicles report	Vehicle identification, type, capacity, mileage and availability are maintained in a structured form.
Maintain driver information	Drivers form and All_Drivers report	Driver details, licence information and availability are stored in the application.
Manage delivery requests	Delivery_Requests form and Pending_Requests report	Delivery requests are recorded with source, destination, load weight, priority and status.
Automate vehicle and driver allocation	Auto_Vehicle_Allocation workflow (DS-01)	An available vehicle with sufficient capacity and an available driver are identified, and a trip record is created.
Calculate trip expenditure	Trip cost calculation workflow	Total trip cost is calculated using fuel and additional expenses. A test value of ₹2,300 was obtained.
Calculate fuel efficiency	Fuel Efficiency Calculation workflow (DS-03)	Distance travelled divided by fuel consumed produced 10 km/L in the test case.
Calculate cost per kilometre	Cost Per KM Calculation workflow (DS-05)	A total cost of ₹2,300 over 200 km produced ₹11.50 per km.
Provide fleet operation summaries	Fleet_Dashboard page	Four summary cards and two pie charts display fleet and delivery information.
Control application access	Admin_Access, Manager_Access and Driver_Access profiles	Access is organised according to the configured application roles.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Vehicle record management	Vehicle information maintained through separate manual records	Vehicles form and All_Vehicles report	Centralised vehicle information.
Driver record management	Driver details maintained separately	Drivers form and All_Drivers report	Structured driver information and availability status.

Parameter	Existing System	Proposed Application	Improvement
Delivery request processing	Manual checking of vehicle capacity and driver availability	Delivery_Requests form with automatic allocation workflow	Reduces manual resource selection.
Vehicle allocation	Vehicle selection performed manually based on capacity	Deluge workflow identifies an available vehicle with sufficient capacity	Automates the resource selection process.
Trip expense calculation	Manual addition of fuel and additional expenses	Trip cost calculation workflow	Reduces repeated manual calculations.
Fuel efficiency calculation	Manual division of distance by fuel consumed	Automatic calculation through Deluge	Provides a calculated fuel efficiency value.
Fleet monitoring	Reviewing separate records to understand fleet status	Fleet_Dashboard with summary cards and charts	Provides a consolidated operational overview.
Access management	Information shared without a structured application-level role setup	Three configured permission profiles	Organises application access according to user roles.

8.4 Limitations of the Low-Code Approach

The application depends on Zoho Creator's platform, data model and available configuration options. Although the platform simplifies the creation of forms, reports, pages and workflows, the design and behaviour of the application are influenced by the features supported by the platform.

Deluge scripting provides flexibility for automating allocation and calculations, but more complex processing may require additional scripting or external services.

The current allocation process uses predefined conditions based on vehicle capacity and availability. It does not perform advanced route optimisation, predictive demand analysis or artificial intelligence-based allocation.

The application also depends on the availability of the Zoho Creator platform and internet connectivity for accessing its published environment.

The project focuses on core fleet operations and trip management. Features such as live GPS tracking, external logistics integrations and advanced route planning are outside the scope of the current implementation.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The Fleet Operations and Vehicle Allocation System demonstrates the use of low-code application development through Zoho Creator.

The forms, lookup fields, reports, dashboard cards and charts represent the visual configuration capabilities of the platform. These components allow application functionality to be created without developing every interface element through conventional programming.

Deluge scripting represents the low-code layer of the project. It is used to implement automatic vehicle and driver allocation, trip cost calculation, fuel efficiency calculation and cost-per-kilometre calculation. These workflows introduce application logic while continuing to use the built-in Zoho Creator environment.

No separate pro-code extension is used in the current implementation. A pro-code approach could be considered in the future if the application requires advanced route optimisation, specialised integrations or processing beyond the available low-code capabilities.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Fleet Operations and Vehicle Allocation System was developed using Zoho Creator to organise vehicle information, driver records, delivery requests and trip details within a centralised application. The application provides separate forms and reports for maintaining fleet resources and monitoring delivery operations.

The Fleet_Dashboard page presents important operational information through four summary cards and two pie charts. The application uses Deluge workflows to automate vehicle and driver allocation based on vehicle capacity and resource availability. When suitable resources are found, the workflow creates a trip record and updates the corresponding allocation statuses.

The Trips form maintains trip information and includes subforms for fuel details and additional expenses. Configured workflows calculate total trip cost, fuel efficiency and cost per kilometre. Testing demonstrated automatic allocation for delivery request R004, a total trip cost of ₹2,300, fuel efficiency of 10 km/L and a cost per kilometre of ₹11.50.

The application also uses three permission profiles to organise access for administrators, managers and drivers. The completed application was published through Zoho Creator.

Overall, the project demonstrates how a low-code platform can be used to develop a structured fleet management application with automated resource allocation, trip calculations, reporting and dashboard-based monitoring.

9.2 Future Enhancements

1. **Advanced Vehicle Allocation:** Extend the existing allocation workflow to consider additional factors such as vehicle mileage, delivery priority and distance when selecting resources.
2. **Route Optimisation:** Introduce route-planning capabilities to identify efficient delivery routes and reduce unnecessary travel distance.

3. **GPS Tracking:** Integrate a suitable location-tracking service to monitor vehicle movement and provide real-time trip information.
4. **Predictive Maintenance:** Introduce vehicle usage and maintenance records to help identify vehicles that may require servicing.
5. **Advanced Analytics:** Extend the dashboard with historical trends, vehicle utilisation reports and delivery performance analysis.
6. **External Integration:** Consider integration with suitable logistics or mapping services if additional operational requirements arise.

9.3 Learning Outcomes and Career Relevance

This project provided practical experience in developing a low-code application using Zoho Creator. It involved designing forms, defining field properties, establishing lookup relationships and organising application records through reports.

The implementation also provided experience in writing Deluge scripts and configuring form workflows for automatic resource allocation and trip-related calculations. Testing the application helped develop an understanding of workflow triggers, field link names, data validation and calculation accuracy.

The project also highlighted the importance of user permissions, dashboard design and structured data management. These skills are relevant to low-code application development, business process automation, software development and application support.

Further practice with Deluge scripting, application testing and platform integrations can help extend the system to support more advanced fleet management requirements.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and sections/menu organised	Yes	Vehicles, Drivers, Delivery_Requests, Trips and Fleet_Dashboard	5.1	Fig. 5.1
II	Forms with multiple field types	Yes	Vehicles, Drivers, Delivery_Requests and Trips forms	5.2	Fig. 5.2, 5.3
II	Lookup field relationships between forms	Yes	Trips to Vehicles, Drivers and Delivery_Requests	4.2, 5.2	Fig. 5.3
II	Field validations and dropdown values	Yes	Main application forms	4.3, 5.2	Table 4.2, 4.3
II	List reports	Yes	All_Vehicles, All_Drivers, Pending_Requests and Trips	5.3	Fig. 5.4
II	Page / dashboard built with UI Builder	Yes	Fleet_Dashboard	5.4	Fig. 5.5
II	Deluge – form event workflow	Yes	Delivery_Requests – Auto_Vehicle_Allocation	5.5	Code 5.1
II	Deluge – record fetch and conditional logic	Yes	Automatic vehicle and driver allocation	5.5	Code 5.1
II	Deluge – arithmetic calculations	Yes	Trip cost, fuel efficiency and cost per kilometre	5.5	Code 5.2, 5.3
II	Subforms	Yes	Fuel_Details and Additional_Expenses in Trips	4.3, 5.2	Fig. 5.3
II	Form workflows	Yes	Allocation and trip calculation workflows	5.6	Fig. 5.6, 5.7
II	Role-based permission profiles	Yes	Admin_Access, Manager_Access and Driver_Access	4.6, 5.8	Fig. 5.9
II	Testing and debugging evidence	Yes	Functional testing and defect log	6	Table 6.1, 6.2
III	Analytics through dashboard charts	Yes	Fleet_Dashboard	5.4, 5.9	Fig. 5.5, 5.10
V	Low-code limitations and spectrum analysis	Yes	Report analysis	8.4, 8.5	–

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture and business value of low-code application development using Zoho Creator.	Ch. 1, 2	Good	
CO2	Develop business applications using Zoho Creator forms, reports, workflows and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	Good	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections and role-based security.	Ch. 4.6, 5.7–5.8	Satisfactory	
CO4	Deploy, secure and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	Good	
CO5	Evaluate low-code development strategies, AI-assisted application development and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	Satisfactory	

Self-assessment scale: Excellent, Good, Satisfactory or Needs Improvement. The Examiner Level column is left blank for the examiner.

Basis for the self-assessment:

- **CO1 – Good:** The principles, architecture and justification for low-code development are covered in Chapters 1 and 2.
- **CO2 – Good:** The project includes multiple forms, lookup relationships, reports, dashboard components, Deluge workflows and application testing.
- **CO3 – Satisfactory:** The application uses internal lookup relationships and configured permission profiles. No external API or third-party integration is included.
- **CO4 – Good:** The application was published through Zoho Creator, and the report covers permission profiles, application maintenance and deployment practices.
- **CO5 – Satisfactory:** The report analyses low-code capabilities and limitations and discusses the transition from visual configuration to Deluge scripting. No AI-based or separate pro-code component is used.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Auto_Vehicle_Allocation (DS-01)

```
// Script ID : DS-01
// Workflow  : Auto_Vehicle_Allocation
// Trigger   : Delivery_Requests -> Created -> Successful form submission

availableVehicles = Vehicles[Availability == "Available" && Capacity_KG >=
input.Load_Weight_KG];
availableDrivers = Drivers[Availability == "Available"];

if(availableVehicles.count(ID) > 0)
{
    if(availableDrivers.count(ID) > 0)
    {
        vehicleRec = availableVehicles;
        driverRec = availableDrivers;

        insert into Trips
        [
            Delivery_Requests = input.ID
            Vehicles = vehicleRec.ID
            Drivers = driverRec.ID
            Trip_Status = "Assigned"
        ];

        vehicleRec.Availability = "Assigned";
        driverRec.Availability = "Assigned";
        input.Request_Status = "Assigned";
    }
    else
    {
        info "No available driver found.";
    }
}
else
{
    info "No suitable vehicle found.";
}
```

C.2 Fuel Efficiency Calculation (DS-03)

```
// Script ID : DS-03
// Purpose   : Calculate fuel efficiency in km/L

if(input.Fuel_Used_L != null && input.Fuel_Used_L > 0)
{
    tripRecord = Trips[ID == input.ID];
    tripRecord.Fuel_Efficiency = input.Distance_KM / input.Fuel_Used_L;
}
else
{
    info "Enter valid fuel usage to calculate fuel efficiency.";
}
```

C.3 Cost Per KM Calculation (DS-05)

```
// Script ID : DS-05
// Purpose   : Calculate cost per kilometre

tripRecord = Trips[ID == input.ID];

if(input.Distance_KM != null && input.Distance_KM > 0)
{
    tripRecord.Cost_Per_KM = tripRecord.Total_Cost / input.Distance_KM;
}
```

C.4 Trip Cost Calculation (DS-02)

Insert the exact final Deluge script saved in the Trip Cost Calculation workflow here. The script should be copied from the Zoho Creator workflow editor so that the submitted report contains the actual implemented version.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

The Fleet Operations and Vehicle Allocation System is a Zoho Creator application. It can be accessed through a compatible web browser using an authorised Zoho Creator account.

1. **Open the application.** Start Google Chrome, Microsoft Edge or Mozilla Firefox and navigate to the published application link.
2. **Sign in.** Log in using the Zoho account that has been granted access to the application.
3. **Open the application.** Select Fleet Operations and Vehicle Allocation System from the available applications.
4. **Navigate through the application.** Use the available sections to access Vehicles, Drivers, Delivery_Requests, Trips and Fleet_Dashboard.
5. **Sign out.** After completing the required operations, sign out of the Zoho account when using a shared device.

D.2 Instructions for the Administrator

1. **Manage vehicles.** Open the Vehicles form to add or update vehicle details such as Vehicle_ID, Vehicle_Name, Vehicle_Type, Capacity_KG, Mileage_KM_L and Availability.
2. **Manage drivers.** Open the Drivers form to add or update driver name, phone number, licence number and availability.
3. **Review delivery requests.** Open the Delivery_Requests form or Pending_Requests report to review delivery requirements.
4. **Monitor allocation.** After a delivery request is submitted, review the Trips report to check whether the allocation workflow has created the corresponding trip record.
5. **Review trip details.** Open the Trips form to view the allocated vehicle and driver and maintain distance, fuel usage and trip status.
6. **Review expenses.** Enter fuel purchase details through Fuel_Details and other trip-related charges through Additional_Expenses.

7. **Monitor calculations.** Check Total_Cost, Fuel_Efficiency and Cost_Per_KM in the Trips record.
8. **Review the dashboard.** Open Fleet_Dashboard to view total vehicles, available vehicles, pending deliveries and completed trips.
9. **Manage access.** Review Admin_Access, Manager_Access and Driver_Access profiles and assign users the permissions required for their responsibilities.
10. **Maintain the application.** Review records periodically, test workflow changes and retain a backup before making major modifications.

D.3 Instructions for the Manager

1. Open the Vehicles report to review vehicle records and their availability.
2. Open the Drivers report to review driver information and availability.
3. Open Pending_Requests to identify delivery requests awaiting allocation.
4. Create a delivery request by entering the request ID, source, destination, load weight and priority in the Delivery_Requests form.
5. After submitting the request, open the Trips report to check the automatically created trip record.
6. Open the Trips form to review trip details and enter the distance travelled and fuel consumed.
7. Record fuel purchases through Fuel_Details and additional expenses through Additional_Expenses.
8. Review the calculated trip cost, fuel efficiency and cost per kilometre.
9. Open Fleet_Dashboard to monitor fleet availability, delivery request status and completed trips.

D.4 Instructions for the Driver

1. Sign in using the authorised Zoho Creator account.
2. Open the Fleet Operations and Vehicle Allocation System.
3. Access the Trips section according to the permissions assigned through Driver_Access.
4. View the trip records available to the account.
5. Update permitted trip information where required.

6. Enter trip details such as distance travelled and fuel consumed when permitted.
7. View the Fleet_Dashboard page to review the available fleet summary information.
8. Sign out after completing the required operations.